Financial Mathematics

Pre-requisite : Probability Theory

Books

1. S. Roman. Introduction to the Mathematics of Finance : From Risk Management to Options Pricing. Springer
2. K. Bosovkov. Elements of Stochastic Modelling Ch 1, 4, 5, 6, 11, 12, 13
3. D. A. Dubofsky and T. W. Miller. Derivatives, Valuation and Risk Management.

		Marks	
Mid Sem		30	
End Sem		30	
TA	Class test I	$20 = \max[I, II]$	40
	Class test II		
	Project	10	
	Atten, Suptal etc	10	
Total		100	

No extra class test if one missed CT-1, CT-2

Review of Probability

Def^n Ω non-empty set. $\mathcal{F}$ is σ-field on Ω is a family of subsets of Ω s.t.

(i) $\phi \in \mathcal{F}$ (ii) If $A \in \mathcal{F} \Rightarrow \Omega - A = A^c \in \mathcal{F}$

(iii) If $A_1, A_2, \ldots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$

Example (i) Toss a coin infinitely many times

$\Omega = \{ [HH\cdots], [HT\cdots], [TT\cdots], [TH\cdots] \}$

$\mathcal{F}_0 = \{\emptyset, \Omega\}$

$\Omega_1 = \{A_H, A_T\}$

$\mathcal{F}_1 = \{\emptyset, A_H, A_T, \Omega_1\}$

$\Omega_2 = \{A_{HH}, A_{HT}, A_{TH}, A_{TT}\}$

$\mathcal{F}_2 = \{\emptyset, A_{HH}, A_{HT}, A_{TH}, A_{TT}, [A_{HH}, A_{HT}], \cdots, \Omega_2\}$

$\vdots$

$\mathcal{F}_0 \subset \mathcal{F}_1 \subset \mathcal{F}_2 \subset \cdots$

$\mathcal{F}_k$ represent all the information available by time k.

Such increasing families of σ-fields are called flows of σ-algebra or filtration.

(ii) $\mathbb{R}$

family of Borel set. $\mathcal{F} = \mathcal{B}(\mathbb{R})$ is a σ-field on $\mathbb{R}$, where $\mathcal{B}(\mathbb{R})$ is the smallest σ-field containing all intervals of $\mathbb{R}$.

Def^n $\mathcal{F}$ is a σ-field on Ω. A prob measure is a fn $P: \mathcal{F} \to [0,1]$ s.t

(i) $P(\Omega) = 1$, $\Omega \in \mathcal{F}$

(ii) If $A_1, A_2, \cdots \in \mathcal{F}$ are pairwise disjoint set (i.e. $A_i \cap A_j = \emptyset$ for $i \neq j$), then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

— x —

$(\Omega, \mathcal{F}, P)$ prob. space

→ The sets belonging to $\mathcal{F}$ are called events.

→ An event A is said to occur almost surely (a.s.) whenever $P(A) = 1$

Ex. 1) If $A_1 \subset A_2 \subset \cdots$, then $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{n\to\infty} P(A_i)$ // $P\left(\lim_{n\to\infty} A_i\right)$

*1 2) If $A_1 \supset A_2 \supset \cdots$, then $P\left(\bigcap_{i=1}^{\infty} A_i\right) = \lim_{n\to\infty} P(A_n)$ // $P\left(\lim_{n\to\infty} A_i\right)$

Lemma (Borel-Cantelli) Let $A_1, A_2, \ldots$ be a sequence of event st $\sum_{i=1}^{\infty} P(A_i) < \infty$ and let $B_n = A_n \cup A_{n+1} \cup \cdots$

Then $P(B_1 \cap B_2 \cap \cdots) = 0$

Pf: $B_n \downarrow$

$$P\left(\bigcap_{n=1}^{\infty} B_n\right) = \lim_{n\to\infty} P(B_n) \quad *_1$$

$$= \lim_{n\to\infty} P(A_n \cup A_{n+1} \cup \cdots)$$

$$\leq \lim_{n\to\infty} \sum_{i=n}^{\infty} P(A_i)$$

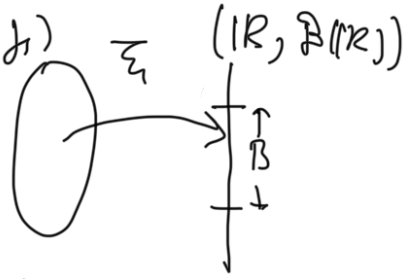
$$= 0,$$

Since the series $\sum_{n=1}^{\infty} P(A_n)$ is convergent

— x —

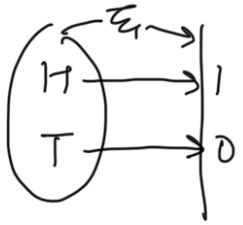
Def^n $(\Omega, \mathcal{F})$, then $\xi : \Omega \to \mathbb{R}$ is said to be $\mathcal{F}$-mble if $[\xi \in B] \in \mathcal{F} \quad \forall\ B \in \mathcal{B}(\mathbb{R})$,

where $[\xi \in B] = \{\omega \in \Omega : \xi(\omega) \in B\} = \xi^{-1}(B)$



→ If $(\Omega, \mathcal{F}, P)$ is a prob. space, then such a fn ξ is called random variable (r.v.)

Eg $\Omega = \{H, T\}$, $\mathcal{F} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$



ξ # of heads $\in \{0, 1\}$

$$\xi^{-1}(-\infty, x] = \begin{cases} \emptyset & , x < 0 \\ \{T\} & , 0 \le x < 1 \\ \{H, T\} & , x \ge 1 \end{cases} \in \mathcal{F}$$

$P(\xi = 0) = P_\xi\{0\} = P(T)$

ξ r.v.

Def^n σ-field $\sigma(\xi)$ generated by a r.v. $\xi : \Omega \to \mathbb{R}$ consists of all sets of the form $[\xi \in B]$, B any a Borel set in $\mathbb{R}$.

Def^n The σ-field $\sigma(\xi_1, \dots, \xi_n)$ [illegible]

Def: The σ-field $\sigma\{\xi_i, i \in I\}$ generated by a family $\{\xi_i, i \in I\}$ of r.v. is defined to be the smallest σ-field containing all events of the form $\{\xi_i \in B\}$, where $B \in \mathcal{B}(\mathbb{R})$, $i \in I$.

→ We call $f : \mathbb{R} \to \mathbb{R}$ a Borel fn if the inverse image $f^{-1}(B) \in \mathcal{B}$, $B \in \mathcal{B}(\mathbb{R})$.

Ex. Show that if f is a Borel fn and ξ is a r.v., then the composition $f(\xi)$ is $\sigma(\xi)$ mble.

Sol If $B \in \mathcal{B}$; $f : \mathbb{R} \to \mathbb{R}$, $f^{-1}(B) \in \mathcal{B}$, $B \in \mathcal{B}$

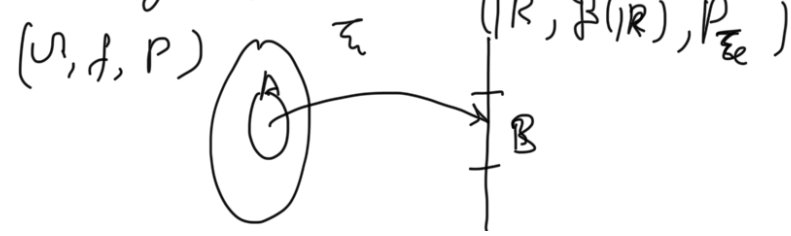
$\therefore \{f(\xi) \in B\} = \{\xi \in f^{-1}(B)\} \in \sigma(\xi)$

Lemma (Doob – Dynkin)

Let ξ be a r.v., Then each $\sigma(\xi)$ mble r.v. η can be written as $\eta = f(\xi)$ for some Borel fn $f : \mathbb{R} \to \mathbb{R}$.

Def: Every r.v. $\xi : \Omega \to \mathbb{R}$ gives rise to a prob. measure

$P_\xi(B) = P(\xi \in B)$ on $\mathbb{R}$ defined on the σ-field of Borel sets $B \in \mathcal{B}(\mathbb{R})$. We call P_ξ the distr of ξ.



→ The fn $F_\xi : \mathbb{R} \to [0,1]$ defined by $F_\xi(x) = P(\xi \le x)$ is called distribution fn (d.f.) of ξ.

Ex. Show that the d.f F_ξ is non-decreasing, right cont. and

$$\lim_{x \to -\infty} F_\xi(x) = 0 \quad , \quad \lim_{x \to +\infty} F_\xi(x) = 1$$

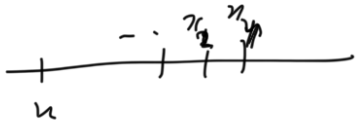
$x_1 < x_2$

$(-\infty, x_1] \subseteq (-\infty, x_2]$

$P_\xi(-\infty, x_1] \le P_\xi(-\infty, x_2]$

$P(\xi \le x_1) \le P(\xi \le x_2)$

$F_\xi(x_1) \le F_\xi(x_2)$



Def^n If there is a Borel fn $f_\xi : \mathbb{R} \to \mathbb{R}$ st. for any Borel set $B \subset \mathbb{R}$

$$P(\xi \in B) = \int_B f_\xi(x)\, dx$$

then ξ is r.v. with absolutely cont. distn and f_ξ is called the density of ξ.

→ If there is (finite or infinite) seq. of pairwise distinct real numbers $x_1, x_2, \ldots$ st. for any $B \in \mathcal{B}(\mathbb{R})$

$$P(\xi \in B) = \sum_{x_i \in B} P(\xi = x_i)$$

then ξ discrete distn with values $x_1, x_2, \ldots$

and mass $P(\xi_i = x_i)$ at x_i.

Def^n Let $\xi_1, \dots, \xi_n$ be

The joint dist^n of

$P_{\xi_1, \dots, \xi_n}$ on $\mathbb{R}^n$ s.t.

$P_{\xi_1, \dots, \xi_n}(B) = P[(\xi_1, \dots, \xi_n) \in B]$ for any Borel set B in $\mathbb{R}^n$.

Borel fn $f_{\xi_1, \dots, \xi_n} : \mathbb{R}^n \to \mathbb{R}$ s.t

$$P((\xi_1, \dots, \xi_n) \in B) = \int_B f_{\xi_1, \dots, \xi_n}(x_1, \dots, x_n)\, dx_1 \cdots dx_n$$

, $B \in \mathcal{B}(\mathbb{R}^n)$

$f_{\xi_1, \dots, \xi_n}$ joint density of $\xi_1, \dots, \xi_n$

Def^n A r.v. $\xi : \Omega \to \mathbb{R}$ is said to be integrable if $\int_\Omega |\xi|\, dP < \infty$.

→ Then $E(\xi) = \int_\Omega \xi\, dP$ exist and is called expectation of ξ.

The family of integrable r.v. $\xi : \Omega \to \mathbb{R}$ will be denoted by L^1 or $L^1(\Omega, \mathcal{F}, P)$

Example $1_A = \begin{cases} 1 & \text{on } A \\ 0 & \text{on } \Omega - A \end{cases}$ or $1_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \in \Omega - A \end{cases}$

For any event A

$$E(1_A) = \int_\Omega 1_A \, dP = P(A)$$

We say that $\eta : \Omega \to \mathbb{R}$ is a step fn if $\eta = \sum_{i=1}^{n} n_i \, 1_{A_i}$,

where $n_1, \dots, n_n$ are real no's and

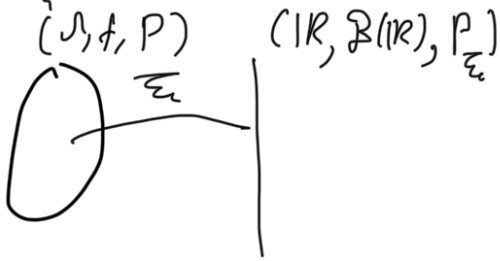
$A_1, \dots, A_n$ are pairwise disjoint events

$$E(\eta) = \int_\Omega \eta \, dP = \sum_{i=1}^{n} n_i \int_\Omega 1_{A_i} dP = \sum_{i=1}^{n} n_i \, P(A_i)$$

Thm : Show that for any Borel fn $h : \mathbb{R} \to \mathbb{R}$

s.t. $h(\xi)$ is integrable, then

$$E(h(\xi)) = \int_{\mathbb{R}} h(x) \, dP_\xi(x)$$



Sol

If h is step fn $h = \sum_{i=1}^{n} h_i \, 1_{A_i}$

real no's

A_i's pairwise disjoint Borel sets covering $\mathbb{R}$.

$$E(h(\xi)) = \sum_{i=1}^{n} h_i \, E(1_{A_i}(\xi))$$

$$= \sum_{i=1}^{n} h_i \, P(\xi \in A_i)$$

$$= \sum_{i=1}^{n} h_i \, P_\xi(A_i)$$

n

$$= \sum_{i=1}^{\infty} \int_{A_i} h(x)\, dP_{\xi}(x)$$

$$= \int_{\mathbb{R}} h(x)\, dP_{\xi}(x)$$

non-negative Borel fn h can be approximated by a non-decreasing ($\uparrow$) seq of step fns

For such an h the result follows from monotone convergence of integrals

MCT $0 \le X_n \uparrow X$ as $n \to \infty$, Then $E(X_n) \uparrow E(X)$

$h = h^+ - h^-$, $h^+, h^- \ge 0$

$\rightarrow$ ξ abs. cont with density f_{ξ}, then

$$E(h(\xi)) = \int_{-\infty}^{\infty} h(x)\, f_{\xi}(x)\, dx$$

$\rightarrow$ ξ discrete $x_1, x_2, \ldots$

$$E(h(\xi)) = \sum_i h(x_i)\, P(\xi = x_i)$$

—x—

Def^n $A, B \in \mathcal{F}$ s.t. $P(B) \neq 0$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Ex Thm of total prob.

$$P(A) = \sum_{i=1}^{\infty} P(A|B_i)\, P(B_i)$$ for any $A \in \mathcal{F}$ and

any set of pairwise disjoint event $B_1, B_2, \ldots \in \mathcal{F}$ s.t. $\bigcup_{i=1}^{\infty} B_i = \Omega$ and $P(B_i) \neq 0$ for any i

Independence

A and B are indep $\iff P(A \cap B) = P(A)\,P(B)$, $A, B \in \mathcal{F}$

$\iff P(A|B) = P(A)$

$\iff P(B|A) = P(B)$

$A_i \in \mathcal{F}$ $i = 1, \ldots, n$ are indep if

$$P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k})$$

for any indices $1 \le i_1 < i_2 < \cdots < i_k \le n$

Def'n Two rv's ξ and η are called indep if for any Borel sets $A, B \in \mathcal{B}(\mathbb{R})$, the any two event $\{\xi \in A\}$ and $\{\eta \in B\}$ are indep

In general a (finite or infinite) family of rv's are indep if any finite # of rv's from this family are indep.

Def'n Two σ-fields $\mathcal{G}$ and $\mathcal{H}$ contained in $\mathcal{F}$ are indep. if any two event $A \in \mathcal{G}$ and $B \in \mathcal{H}$ are indep.

Ex. Two rv's ξ and η are indep if and only if $\sigma(\xi)$ and $\sigma(\eta)$ generated by them are indep.

Def'n We say that

say that a r.v. X is indep of a σ-field $\mathcal{G}$ if the σ-field $\sigma(X)$ and $\mathcal{G}$ are indep.

—x—

$P(X>t) = E(P(X>t|Y))$ $\quad E(E(X|Y))$

$= \int_{-\infty}^{\infty} \int_{t}^{\infty} f_{X|Y=y}(x)\, dx\, f_Y(y)\, dy$ $\quad \int \left(\int x\, f_{X|Y=y}(x)\, dx \right) f_Y(y)\, dy$

$= \int_{t}^{\infty} \left(\int_{-\infty}^{\infty} f(x,y)\, dy \right) dx$ $\quad = \int \int x\, f_{X|Y=y}(x)\, f_Y(y)\, dx\, dy$

$= \int_{t}^{\infty} f_X(x)\, dx$ $\quad = \int \int x\, f(x,y)\, dx\, dy$

$= P(X>t)$ $\quad = \int x \left(\int f(x,y)\, dy \right) dx$

$= \int x\, f_X(x)\, dx = E(X)$

$\rightarrow P(X>t) = E(P(X>t|Y))$

(1) linearity

$E(aX+bY|\mathcal{G}) = a\,E(X|\mathcal{G}) + b\,E(Y|\mathcal{G})$

(2) If Z is $\mathcal{G}$ mble, then $E(ZX|\mathcal{G}) = Z\,E(X|\mathcal{G})$
$\{Z \in B\} \in \mathcal{G}\ \forall B \in \mathcal{B}(\mathbb{R})$

(3) If X is indep of $\mathcal{G}$ (meaning that $B \in \mathcal{B}$ and $A \in \mathcal{G}$, the ever $\{X \in B\}$ and A are

map), then $E(X|\mathcal{G}) = E(X)$

In particular, if $X \equiv c$ = const or $\mathcal{G} = \{\emptyset, \Omega\}$ is trivial, then we have

$$E(X|\mathcal{G}) = X$$

(4) If $\mathcal{G}_0 \subset \mathcal{G}_1$ ($\mathcal{G}_i$ are σ-fields)

$$E\left(E(X|\mathcal{G}_1)|\mathcal{G}_0\right) = E(X|\mathcal{G}_0)$$

Set $B \in \mathcal{G}_0$

$$E\left(E(X|\mathcal{G}_1)|B\right) = \frac{1}{P(B)} \int_B E(X|\mathcal{G}_1)\, dP$$

$$= \frac{1}{P(B)} \int_B X\, dP$$

$$= E(X|B)$$

CE1. $E(aX+bY|\mathcal{G}) = aE(X|\mathcal{G}) + bE(Y|\mathcal{G})$

CE2 If Z is $\mathcal{G}$mble, then $E(ZX|\mathcal{G}) = ZE(X|\mathcal{G})$

$[Z \in B^*] \in \mathcal{G} \quad \forall B^* \in \mathcal{G}$

CE3 If X is indep of $\mathcal{G}$ (meaning: for any $B \in \mathcal{B}$ and $A \in \mathcal{G}$, the event $[X \in B]$ and A are indep.), then

$$E(X|\mathcal{G}) = E(X)$$

CE4 If $\mathcal{G}_0 \subset \mathcal{G}_1 \subset \mathcal{F}$ are σ-fields, then

$$E\left(E(X|\mathcal{G}_1)|\mathcal{G}_0\right) = E(X|\mathcal{G}_0)$$

Seal $B \in \mathcal{G}_0$

$$E\left(E(X|\mathcal{G}_1)|B\right) = \frac{1}{P(B)} \int_B E(X|\mathcal{G}_1)\,dP$$

$$= \frac{1}{P(B)} \int_B X\,dP$$

$$= E(X|B)$$

—x—

Martingale (MG):

discrete time process $(X_t)_{t \in T}$ on $(\Omega, \mathcal{F}, P)$

$T \subseteq \{0,1,2,\dots,T\}$, $T < \infty$

or $\{0,1,2,\dots\}$ or $\{\dots,-1,0,1,\dots\}$

→ given t, $\mathcal{F}_t$ collection of event "observable" by that time

→ filtration $\underline{\mathbb{F}}$ is ↑ seq. of sub-σ fields

$$\mathcal{F}_0 \subset \mathcal{F}_1 \subset \cdots \subset \mathcal{F}$$

→ sp $(X_t)_{t\geq 0}$ is adapted to filtration $\underline{\mathbb{F}}$ if for any $t = 0, 1, 2, \ldots$ the r.v. X_t is $\mathcal{F}_t$ mble, i.e. $\{X_t \in B\} \in \mathcal{F}_t$ for any $B \in \mathcal{B}$

Also f mble fn, $(Y_t \equiv f(X_t))_{t\geq 0}$ is also adapted

since $\mathcal{F}_t \subset \mathcal{F}_s$, $t \leq s$

$\therefore$ X_t $\mathcal{F}_s$ mble for any $s \geq t$ as well

$(\Omega, \mathcal{F}, \underline{\mathbb{F}}, P)$ Filtered prob. space or stochastic basis

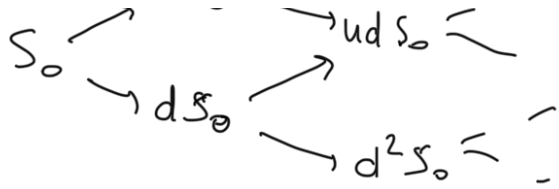
Example Binomial market model

Price S_t of a given asset, S_0 fixed (assume)

for $d < u$

$$S_{t-1} \begin{cases} \nearrow S_t = u S_{t-1} \\ \searrow S_t = d S_{t-1} \end{cases}, \quad t = 1, 2, \ldots, T$$

$$\begin{cases} \nearrow u^2 S_0 \\ u S_0 \end{cases}$$



$t=0 \quad t=1 \quad t=2 \qquad\qquad t=T$

$\Omega := \{ \underset{\sim}{\nu} = (\nu_1 \nu_2 \cdots \nu_T) : \nu_j = d \text{ or } u \}$

$\mathcal{F}_t$

$\mathcal{F}_0 = \{\emptyset, \Omega\}$, $\mathcal{F}_1 := \sigma(A_d, A_u) \equiv \{\emptyset, A_d, A_u, \Omega\}$

$A_d = \{\nu \in \Omega : \nu_1 = d\}$

$\mathcal{F}_2 = \sigma(A_{dd}, A_{du}, A_{ud}, A_{uu})$

$A_{du} = \{\nu \in \Omega : \nu_1 = d, \nu_2 = u\}$

$A_d = A_{dd} \cup A_{du}$

$A_u = A_{ud} \cup A_{uu}$

$\mathcal{F}_0 \subset \mathcal{F}_1 \subset \mathcal{F}_2 \subset \cdots$

S_t is adapted to filtration $\underset{\sim}{\mathcal{F}}$

S_t^2 " " " $\underset{\sim}{\mathcal{F}}$

$Y_t := S_{t+1} - S_t$ is NOT adapted

$\longrightarrow$ $(\Omega, \mathcal{F}, \underset{\sim}{\mathcal{F}}, P)$ discrete time SP $(X_t)_{t \geq 0}$ adapted to filtration $\mathcal{F}$

(X_t) is Martingale (MG) if for any $t = 0, 1, 2, \ldots$

$E|X_t| < \infty$ and $E(X_{t+1} | \mathcal{F}_t) = X_t$

for $s \geq 1$

$E(X_{t+s} | \mathcal{F}_t) = E\left(E(X_{t+s} | \mathcal{F}_{t+s-1}) | \mathcal{F}_t \right)$

$$= E(X_{t+s-1} \mid \mathcal{F}_t) \quad \because |X_t| \text{ is MG}$$

$$= E(X_{t+1} \mid \mathcal{F}_t) = X_t$$

$$\Rightarrow E(X_{t+s}) = E\big(E(X_{t+s} \mid \mathcal{F}_t)\big) = E(X_t) = E(X_0)$$

—x—

→ $|X_t|$ submartingale (Sub-MG) if:

$E|X_t| < \infty$ and $E(X_{t+1} \mid \mathcal{F}_t) \geq X_t$

, $t = 0, 1, 2, \ldots$

→ $|X_t|$ supermartingale (super-MG) if

$E|X_t| < \infty$ and $E(X_{t+1} \mid \mathcal{F}_t) \leq X_t$

, $t = 0, 1, 2, \ldots$

→ A cont time MG is defined as an adapted proces $(|\mathcal{F}_t|_{t \geq 0}$ $s, t \geq 0$, $\mathcal{F}_t \subset \mathcal{F}_{t+s} \subset \mathcal{F})$

$|X_t|_{t \geq 0}$ st for any $s, t \geq 0$

$E|X_t| < \infty$ and $E(X_{t+s} \mid \mathcal{F}_t) = X_t$

→ SP collection of rv $\xi(t)$

(Stochastic process) $\xi(t) \in S$ ← State space

$t \in T$ ← parameter space/time space

→ For every $\omega \in \Omega$ the fn

$T \ni t \longmapsto \xi(t, \omega)$

is called

a path (or sample path) of $\xi(t)$

→ Wiener process / Standard Brownian Motion (SBM)

A S.P $W(t)$, $t \geq 0$ is a Wiener process iff the following conditions holds

(1) $W(0) = 0$ a.s.

(2) the sample path $t \longmapsto W(t)$ are continuous a.s

(3) $W(t)$ has stationary and independent increment

(4) the increment $W(t) - W(s)$ has normal distn with mean 0 and var. $t-s$ for any $0 \leq s < t$.

Ex SP's are MG wrt natural filteration of SBM $(W_t)_{t \geq 0}$

(i) $(W_t)_{t \geq 0}$

(ii) $(Y_t = W_t^2 - t)_{t \geq 0}$

(iii) $(Z_t := e^{uW_t - u^2 t/2})$

Sol (i) $E(W_t^2) = t$ | $(E(\xi\eta))^2 \leq E(\xi^2)E(\eta^2)$

$(E(W_t))^2 \leq E(W_t^2) = t < \infty$ $\quad (E(\xi_1))^2 \leq E(\xi^2)$

$\Rightarrow E|W_t| < \infty$

$t > s$

$$E(W_t | \mathcal{F}_s) = E(W_t - W_s + W_s | \mathcal{F}_s)$$

$$= E(W_t - W_s | \mathcal{F}_s) + E(W_s | \mathcal{F}_s)$$

$$= E(W_t - W_s) + W_s$$

$$= W_s$$

$\therefore$ W_t is a MG.

(ii) $Y_t = W_t^2 - t$

$$E|Y_t| \le \underbrace{E(W_t^2)}_{=t} + t = 2t < \infty$$

$s, t \ge 0$

$$E(Y_{t+s} | \mathcal{F}_t) = E(W_{t+s}^2 - (t+s) | \mathcal{F}_t)$$

$$= E((W_{t+s} - W_t + W_t)^2 | \mathcal{F}_t) - (t+s)$$

$$= \underbrace{E((W_{t+s} - W_t)^2 | \mathcal{F}_t)}_{=s} + \underbrace{E(W_t^2 | \mathcal{F}_t)}_{=W_t^2} + 2\underbrace{E(W_t(W_{t+s} - W_t) | \mathcal{F}_t)}_{=2W_t E(W_{t+s} - W_t | \mathcal{F}_t) = 0} - (t+s)$$

$$= s + W_t^2 + 0 - t - s$$

$$= W_t^2 - t = Y_t$$

$\therefore$ $Y_t = W_t^2 - t$ is a MG

(iii) $Z_t = e^{uW_t - u^2 t/2}$

$$E|Z_t| = E(Z_t) = e^{-u^2 t/2} \underbrace{E(e^{uW_t})}_{= e^{\frac{1}{2}tu^2}} = 1 < \infty$$

$$E\left(\frac{Z_{t+s}}{Z_t}\,\middle|\,\mathcal{F}_t\right) = E\left(\exp\left\{u(W_{t+s}-W_t)-\frac{u^2}{2}(t+s-t)\right\}\middle|\mathcal{F}_t\right)$$

$$= E\left(e^{u(W_{t+s}-W_t)}\middle|\mathcal{F}_t\right)\cdot e^{-u^2 s/2}$$

$$= E\left(e^{u(W_{t+s}-W_t)}\right)\; e^{-u^2 s/2}$$

$$= E\left(e^{uW_s}\right)\; e^{-u^2 s/2}$$

$$= e^{\frac{1}{2}su^2}\, e^{\frac{1}{2}su^2}$$

$$= 1$$

$$\Rightarrow \quad E(Z_{t+s}\,|\,\mathcal{F}_t) = Z_t$$

$\therefore$ $\{Z_t\}$ is a MG.

Ex. (2) Random walk

$X_0 := 0$, $X_n := Y_1 + \cdots + Y_n$, $n \geq 1$

Y_j's IID rv with $E|Y_1| < \infty$

Whether the sp $\{X_n\}_{n\geq 0}$ is MG?

Sol $\quad E|X_n| \leq \sum_{i=1}^{n} E|Y_i| < \infty$

$$E(X_{n+1}\,|\,\mathcal{F}_n) = E(X_n + Y_{n+1}\,|\,\mathcal{F}_n)$$

$$= E(X_n\,|\,\mathcal{F}_n) + E(Y_{n+1}\,|\,\mathcal{F}_n)$$

$$= X_n + E(Y_{n+1})$$

$$= X_n + E(Y_1)$$

$\therefore$ X_n is an MG if $E(Y_1) = 0$

Ex. Geometric Random walk

$$X_n = X_0 e^{Y_1 + \cdots + Y_n}, \quad n \geq 1$$

When $X_0 := \text{const} > 0$, Y_j's IID rv's. When is $(X_n)_{n \geq 0}$ a MG wrt filtration $\mathcal{F}_n := \sigma(Y_1, \ldots, Y_n)$ ($\mathcal{F}_0$ being trivial

Stopping time (ST)

$(\Omega, \mathcal{F}, \mathbb{F}, P)$ filtered prob. space

r.v. τ on that space is called ST if one has $[\tau \le t] \in \mathcal{F}_t$ for each $t = 0, 1, 2, \ldots$ — (1)

— x —

For a ST τ

$$[\tau = t] = \underbrace{[\tau \le t]}_{\in \mathcal{F}_t} \cap \underbrace{[\tau \le t-1]^c}_{\in \mathcal{F}_{t-1} \subset \mathcal{F}_t} \in \mathcal{F}_t \rightarrow (2)$$

$$[\tau \le t] = \bigcup_{s=0}^{t} \underbrace{[\tau = s]}_{\in \mathcal{F}_s \subset \mathcal{F}_t} \in \mathcal{F}_t$$

(2) $\Rightarrow$ (1)

Eg Let ξ_n be a seq. of r.v.'s adapted to filtration $\mathcal{F}_n$ and let $B \subseteq \mathbb{R}$ be a Borel set. Show that the time of entry of ξ_n into B, i.e.,

$$\tau = \min\{n : \xi_n \in B\}$$ is a S.T.

Sol If $\tau = \min\{n : \xi_n \in B\}$

for any n

$$[\tau = n] = \underbrace{[\xi_1 \notin B]}_{\in \mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)} \cap \cdots \cap \underbrace{[\xi_{n-1} \notin B]}_{\in \mathcal{F}_n} \cap \underbrace{[\xi_n \in B]}_{\in \mathcal{F}_n} \in \mathcal{F}_n$$

$\therefore \tau$ is S.T.

Ex (First hitting time) adapted process $\{X_t\}$, a (boundary) fn u_t, $t = 0, 1, 2, \ldots$

Show that the first hitting (or crossing) time $\tau := \inf\{t \geq 0 : X_t \geq u_t\}$ is a S.T.

— x —

Can one win certainly or at least on av. using a cleverly chosen ST?

ST τ s.t. $E(X_\tau) > E(X_0)$?

No.

Thm: $\{X_t\}$ MG, τ S.T. on $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, P)$.

$Z_t := X_{t \wedge \tau}$, $t = 0, 1, 2, \ldots$ is MG on $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, P)$ $\quad a \wedge b = \min(a, b)$

Sol $Z_0 = X_0$

$t \geq 0$ $\quad Z_{t+1} = \sum_{k=0}^{t} X_k 1_{\{\tau = k\}} + X_{t+1} 1_{\{\tau > t\}}$

$E|Z_{t+1}| \leq E\left(\sum_{k=0}^{t+1} |X_k|\right) = \sum_{k=0}^{t+1} E|X_k| < \infty$

$\because X_t$ is MG

$E(Z_{t+1} | \mathcal{F}_t) = E\left(\sum_{k=0}^{t} X_k 1_{\{\tau = k\}} + X_{t+1} 1_{\{\tau > t\}} \,\middle|\, \mathcal{F}_t\right)$

$= \sum_{k=0}^{t} X_k 1_{\{\tau = k\}} + E(X_{t+1} 1_{\{\tau > t\}} | \mathcal{F}_t)$

$= 1_{\{\tau > t\}} E(X_{t+1} | \mathcal{F}_t)$

$= 1 \quad X_t$

$$= \sum_{k=0}^{t} X_k 1_{(\tau=k)} + X_t 1_{(\tau>t)} \quad \left[\because X_t \text{ is MG}\right.$$

$$= X_{t\wedge\tau} = Z_t$$

$\therefore Z_t$ is MG

Thm! Optimal stopping thm / Martingale stopping time

$(X_t)_{t\geq 0}$ MG, τ bdded ST (i.e. for a const $c < \infty$, we has $\tau < c$ a.s.)

Then $E(X_\tau) = E(X_0)$

(Thus, in a fair game, we can not invent a rule for quitting the game that would "beat the system": the game will remain fair.

Pf $Z_t = X_{t\wedge\tau}$ is an MG

$\because$ MG has const mean

$\Rightarrow E(X_{t\wedge\tau}) = E(X_0)$

Setting $t = c$ $\quad c\wedge\tau = \tau$

$$E(X_\tau) = E(X_0)$$

—x—

Example Let $T = \min\{t : W(t) = 2-4t\}$, i.e. T is the first time that SBM hits the line $2-4t$.

Use martingale stopping thm to find $E(T)$

Sol By martingale stopping thm

$E[W(T)] = E[W(0)] = 0$

$E(W(T)) - E(W(0)) = 0$

$W(T) = 2 - 4T$

$\Rightarrow \quad E(2-4T) = 0 \Rightarrow 2 - 4E(T) = 0$

$\Rightarrow \quad E(T) = \frac{1}{2}.$

$$\frac{\partial f}{\partial t} = \frac{1}{2}\frac{\partial^2 f}{\partial x^2} \quad \text{heat eq}^n$$

$$f_{W(t)}(x) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/2t}$$

, CT → 1 2 Sep

Brownian Motion (BM) with variance parameter σ^2

$\{B(t), t \geq 0\}$ BM if

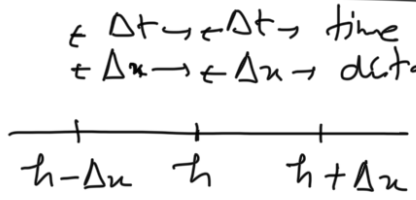
(1) $B(0) = 0$ (2) The sample path $t \longmapsto B(t)$ are continuous a.s

(3) $B(t)$ has stationary & indep increment

(4) $\forall\, t > 0$, $B(t) \sim N(0, \sigma^2 t)$

$B(t) = \sigma W(t)$ $W(t)$ Standard BM / Weiner process

BM can be seen as sum of the random walks



$B(t)$ position of particle at time t

indep ← $X_i = \begin{cases} +1 & \text{if } i^{th} \text{ step +ve direction} \\ -1 & \text{" " " -ve "} \end{cases}$

$B(0) = 0$

$$B(t) = \Delta x \left(X_1 + X_2 + \cdots + X_{\left[\frac{t}{\Delta t}\right]}\right)$$

$P(X_i = +1) = P(X_i = -1) = \frac{1}{2}$

$$E(B(t)) = \Delta x \sum_{i=1}^{[t/\Delta t]} \underbrace{E(X_i)}_{=0} = 0$$

$[t/\Delta t]$

$$V(B(t)) = (\Delta x)^2 \sum_{i=1}^{\quad} \underbrace{V(X_i)}_{=1} = (\Delta x)^2 \left[\frac{t}{\Delta t}\right]$$

If we let $\Delta x = \sigma\sqrt{\Delta t}$ as $\Delta t \to 0$

$$E(B(t)) = 0$$

$$V(B(t)) \longrightarrow \sigma^2 t$$

— x —

$W(t) = \frac{B(t)}{\sigma}$ density $f_t(x) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/2t}$ $-\infty < x < \infty$.

$W(t_1) = x_1,\ W(t_2) = x_2 \ldots \to W(t_n) = x_n$

$\equiv W(t_1) = x_1,\ W(t_2) - W(t_1) = x_2 - x_1, \ldots, W(t_n) - W(t_{n-1}) = x_n - x_{n-1}$

Joint density of $W(t_1), \ldots \to W(t_n)$ is

$$f(x_1, \ldots \to x_n) = f_{t_1}(x_1)\ f_{t_2 - t_1}(x_2 - x_1) \cdots f_{t_n - t_{n-1}}(x_n - x_{n-1})$$

$$= \frac{\exp\left\{-\frac{1}{2}\left\{\frac{x_1^2}{t_1} + \frac{(x_2 - x_1)^2}{t_2 - t_1} + \cdots + \frac{(x_n - x_{n-1})^2}{t_n - t_{n-1}}\right\}\right\}}{(2\pi)^{n/2}\left(t_1(t_2 - t_1)\cdots(t_n - t_{n-1})\right)^{1/2}}$$

$t_1 < t_2 \cdots < t_n$, $x_1 < x_2 \cdots < x_n$

Conditional density of $W(s)$ given $W(t) = B$ for $s < t$

$$f_{s|t}(x|B) = \frac{f_{s,t}(x, B)}{f_t(B)}$$

$$= k_1 \exp\left\{-\frac{1}{2}\left[\frac{x^2}{s} + \frac{(B - x)^2}{t - s}\right]\right\}$$

$\frac{t - s + s}{s(t - s)}$

$$= k_2 \exp\left\{ -\frac{x^2}{2}\left[\frac{1}{s} + \frac{1}{t-s}\right] + \frac{Bx}{t-s} \right\}$$

$$= k_3 \exp\left[-\frac{\left(x - \frac{Bs}{t}\right)^2}{2\, s(t-s)/t} \right]$$

k_1, k_2, k_3 are constants which does not involve x

For $s < t$

$$[W(s) \mid W(t) = B] \sim N\left(\frac{Bs}{t}, \frac{s(t-s)}{t}\right)$$

$s < t$ $\quad E(W(s) \mid W(t) = B) = \frac{Bs}{t}$

$s < t$ $\quad V(W(s) \mid W(t) = B) = \frac{s(t-s)}{t}$

For $s > t$

$$[W(s) \mid W(t) = B] \overset{d}{=} [W(s) - W(t) + W(t) \mid W(t) = B]$$

$$\overset{d}{=} [W(s) - W(t) \mid W(t) = B] + B$$

$$\overset{d}{=} [W(s) - W(t)] + B$$

$$\overset{d}{=} W(s-t) + B$$

$$\sim N(B, s-t)$$

BM with drift $\{B(t), t \geq 0\}$ BM process with drift parameter μ if

(i) $B(0) = 0$ (ii) The sample path $t \mapsto B(t)$ are cont a.s.

(iii) $B(t)$ has stationary & indep increments

(iv) $B(t) \sim N(\mu t, t)$

— x —

SBM $W(t) \sim N(0,t)$

$B(t) = W(t) + \mu t$

$E(X_i) = 2p - 1$

$V(X_i) = E(X_i^2) - [E(X_i)]^2$

$= 1 - (2p-1)^2$

indep $X_i = \begin{cases} +1 & \text{if } i^{th} \text{ step +ve direction} \\ -1 & \text{" " -ve "} \end{cases}$

$P(X_i = +1) = p$

$P(X_i = -1) = 1-p$

$B(t)$: position at time t

$B(t) = \Delta x \left(X_1 + X_2 + \cdots + X_{\left[\frac{t}{\Delta t}\right]} \right)$

eg $[3.2] = 3$

$E(B(t)) = \Delta x \times \left[\frac{t}{\Delta t}\right] \times (2p-1)$

$V(B(t)) = (\Delta x)^2 \times \left[\frac{t}{\Delta t}\right] \times \left(1 - (2p-1)^2\right)$

let $\Delta x = \sqrt{\Delta t}$, $p = \frac{1}{2}\left(1 + \mu\sqrt{\Delta t}\right)$

and let $\Delta t \to 0$

$E(B(t)) \to \mu t$

$V(B(t)) \to t$

$\Delta t \times \frac{t}{\Delta t} \times (1 - \mu^2 \Delta t)$

$\to t$ as $\Delta t \to 0$

Gambler's ruin problem:

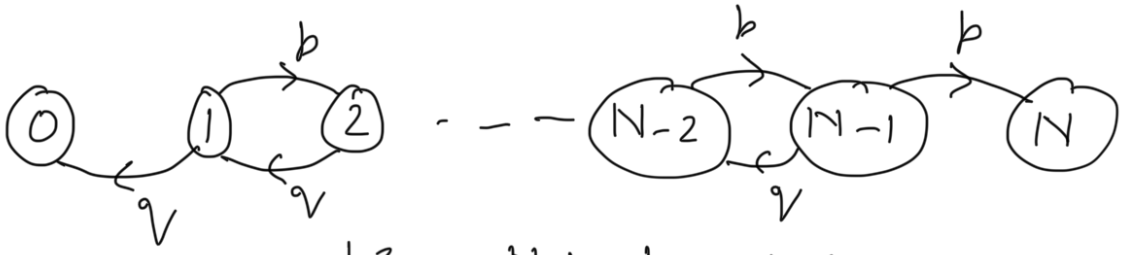
Rs i , $i = 0, 1, \ldots, N$

Aim Rs N

indep Z_i i^{th} bet $P(Z_i = 1) = p$; $P(Z_i = -1) = 1-p = q$

$X_n = i + Z_1 + \cdots + Z_n \in \{0, 1, \ldots, N\}$

Gambler's fortune after n steps



1, 2, --, N-1 transient state

0, N → recurrent state

T_0 : time Gambler broke $= \inf\{n : X_n = 0\}$

T_N : time Gambler has Rs N

$= \inf\{n : X_n = N\}$

$$P_i = P(T_N < T_0)$$

$$P_i = P(T_N < T_0 \mid Z_1 = +1)\, P(Z_1 = +1) + P(T_N < T_0 \mid Z_1 = -1)\, P(Z_1 = -1)$$

$$(p+q)P_i = pP_{i+1} + qP_{i-1}$$

$$P_{i+1} - P_i = \frac{q}{p}(P_i - P_{i-1})$$

$P_0 = 0$, $P_N = 1$

$i=1$, $P_2 - P_1 = \frac{q}{p}(P_1 - P_0) = \frac{q}{p}P_1$

$i=2$, $P_3 - P_2 = \frac{q}{p}(P_2 - P_1) = \left(\frac{q}{p}\right)^2 P_1$

$$P_i - P_{i-1} = \left(\frac{q}{p}\right)^{i-1} P_1$$

$$P_i - P_1 = \left(\frac{q}{p} + \left(\frac{q}{p}\right)^2 + \cdots + \left(\frac{q}{p}\right)^{i-1}\right) P_1$$

$$P_i = \left(1 + \frac{q}{p} + \left(\frac{q}{p}\right)^2 + \cdots + \left(\frac{q}{p}\right)^{i-1}\right) P_1$$

$$= \begin{cases} \dfrac{1-\left(\frac{q}{p}\right)^i}{1-\frac{q}{p}} P_1 & \frac{q}{p} \neq 1 \\ i P_1 & , \quad \frac{q}{p} = 1 \end{cases}$$

$$P_N = 1$$

$$P_i = \begin{cases} \dfrac{1-\left(\frac{q}{p}\right)^i}{1-\left(\frac{q}{p}\right)^N} & , \quad \frac{q}{p} \neq 1 \Leftrightarrow p \neq \frac{1}{2} \\ \dfrac{i}{N} & , \quad \frac{q}{p} = 1 \Leftrightarrow p = \frac{1}{2} \end{cases}$$

If $N \to \infty$

$$P_i = \begin{cases} 0 & , \quad \frac{q}{p} = 1 \Leftrightarrow p = \frac{1}{2} \\ 0 & \frac{q}{p} > 1 \Leftrightarrow 1 > 2p \Leftrightarrow p < \frac{1}{2} \\ 1-\left(\frac{q}{p}\right)^i & \frac{q}{p} < 1 \Leftrightarrow p > \frac{1}{2} \end{cases}$$

$X(t)$ BM with drift $\times$

Probability that the process will hit A before $-B$

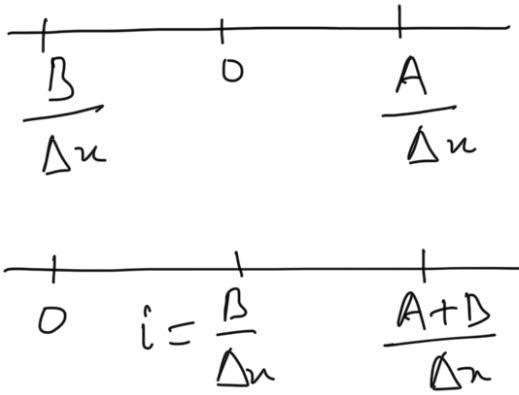
; $A, B > 0$

$$P(x) = P(X(t) \text{ hits } A \text{ before } -B \mid X(0) = x), \quad -B < x < A$$

$$P(0) = P(X(t) \text{ hits } A \text{ before } -B \mid X(0) = 0), \quad -B < 0 < A$$

$$P(A)=1, \quad P(-B)=0$$

$$P(\text{up } A \text{ before down } B) = \frac{1-\left(\frac{1-p}{p}\right)^{B/\Delta x}}{1-\left(\frac{1-p}{p}\right)^{(A+B)/\Delta x}} \qquad (1)$$



$$p = \frac{1}{2}(1+\mu\,\Delta x)$$

$$\lim_{\Delta x \to 0}\left(\frac{1-p}{p}\right)^{1/\Delta x} = \lim_{\Delta x \to 0}\left(\frac{1-\mu\Delta x}{1+\mu\Delta x}\right)^{1/\Delta x}$$

$$= \frac{e^{-\mu}}{e^{\mu}} = e^{-2\mu}$$

$$(1+\mu\Delta x)^{1/\Delta x} = 1 + \frac{1}{\Delta x}\times\mu\Delta x + \frac{\frac{1}{\Delta x}\left(\frac{1}{\Delta x}-1\right)}{2!}(\mu\Delta x)^2 + \cdots$$

Letting $\Delta x \to 0$, we have in (1)

$$P(\text{up } A \text{ before down } B) = \frac{1-e^{-2B\mu}}{1-e^{-2(A+B)\mu}}$$

$$= e^{2\mu A}\,\frac{e^{2B\mu}-1}{e^{2\mu(A+B)}-1} \qquad (2)$$

→ If $\mu < 0$, then from (2), by letting $B \to \infty$

$$P(\text{process ever goes up } A) = e^{2\mu A}$$

$\longrightarrow$ For a BM if we let $\mu \to 0$ in (2), we have

$$P(\text{BM goes up } A \text{ before down } B) = \frac{B}{A+B}$$

In general

$$P(x) = \frac{1 - e^{-2\mu(x+B)}}{1 - e^{-2\mu(A+B)}}$$

— x —

$\mu = 0$

$X(t) = W(t)$ MG

Using MG stopping thm:

$$T = \min\{t ; X(t) = A \text{ or } X(t) = -B\}$$

$$0 = E(X(0)) = E(X(T))$$

$$= A \cdot P(X(T) = A) - B \cdot P(X(T) = -B)$$

$$= A\, P(X(T) = A) - B\,(1 - P(X(T) = A))$$

$$P(X(T) = A) = \frac{B}{A+B} \quad ; \quad P(X(T) = B) = \frac{A}{A+B}.$$

— x —

Geometric BM: (GBM)

$\{Y(t), t \geq 0\}$ BM with drift μ, var. σ^2

$X(t) = e^{Y(t)}$ Geo. BM

— x —

$X(t)$ Geo BM

$s < t$

$E(X(t) \mid X(u),\ 0 \leq u \leq s)$

$= \ldots Y(t)$

$$= E\left(e^{\cdots} \mid Y(u),\ 0 \le u \le s\right)$$

$$= E\left(e^{Y(t)-Y(s)+Y(s)} \mid Y(u),\ 0 \le u \le s\right)$$

$$= e^{Y(s)} E\left(e^{Y(t)-Y(s)} \mid Y(u),\ 0 \le u \le s\right)$$

$$= e^{Y(s)} E\left(e^{Y(t)-Y(s)}\right) \quad \Big| \text{ Using indep increment of } Y(t)$$

$$\Big| \quad W = Y(t-s) \overset{d}{=} Y(t)-Y(s)$$
$$\sim N(\mu(t-s), \sigma^2(t-s))$$
$$E(e^{aW}) = e^{\mu(t-s)a + \frac{1}{2}\sigma^2(t-s)a^2}$$

$$= e^{Y(s)} \cdot e^{\mu(t-s) + \frac{1}{2}\sigma^2(t-s)}$$

$$= X(s)\ e^{(t-s)(\mu + \sigma^2/2)}$$

$$\Rightarrow \quad E(X(t)) = E(X(s)) \cdot e^{(t-s)(\mu+\sigma^2/2)}$$

GBM is useful in modeling stock price over time when you feel the % age changes are IID.

eg X_n : price some stock at time n

assume $\dfrac{X_n}{X_{n-1}}$ are IID

Let $Y_n = \dfrac{X_n}{X_{n-1}}$

$\Rightarrow X_n = Y_n X_{n-1}$

indep $\Big|\Big|$ X_1 and X_2 are indep
$P(X_1 \in B_1, X_2 \in B_2)$
$= P(X_1 \in B_1, X_2 \in B_2)$
$\forall B_1, B_2 \in \mathcal{B}$

$\Big(P(f_1(X_1) \in B_1, f_2(X_2) \in B_2)$

$$= Y_n Y_{n-1} X_{n-2}$$
$$\vdots$$
$$= Y_n Y_{n-1} \cdots Y_1 X_0$$
$$\Rightarrow \log X_n = \sum_{i=1}^{n} \log Y_i + \log X_0$$

$= P(X_1 \in f_1^{-1}(B_1), X_2 \in f_2^{-1}(B_2))$
$= P(X_1 \in f_1^{-1}(B_1)) P(X_2 \in f_2^{-1}(B_2))$
$= P(f_1(X_1) \in B_1)$
$P(f_2(X_2) \in B_2)$
$f_1(X_1), f_2(X_2)$ are indy

Since $\log(Y_i), i \geq 1$ are IID,

$\{\log(X_n)\}$ will when suitably normalized, approx a BM with drift. $\Rightarrow \{X_n\}$ GBM.

—x—

Multivariate Normal dist (MVN)

Let $Z_i \sim NID(0,1)$, $i = 1, \ldots, n$

a_{ij}, $1 \leq i \leq m$, $1 \leq j \leq n$ and μ_i $1 \leq i \leq m$

$$X_i = a_{i1} Z_1 + \cdots + a_{in} Z_n + \mu_i, \quad i = 1, \ldots, m$$

then rvs $X_1, \ldots, X_m$ are said to have MVN.

$$E(X_i) = \mu_i \; ; \; V(X_i) = \sum_{j=1}^{n} a_{ij}^2$$

$$E\left(\sum_{i=1}^{m} t_i X_i\right) = \sum_{i=1}^{m} t_i \mu_i$$

$$V\left(\sum_{i=1}^{m} t_i X_i\right) = \sum_{i=1}^{m} \sum_{j=1}^{m} t_i t_j \, Cov(X_i, X_j)$$

$$Z = \sum_{i=1}^{m} t_i X_i \sim N(-, -)$$

Joint mgf $X_1, \ldots, X_m$

$$\phi(t_1, \ldots, t_m) = E(e^{t_1 X_1 + \cdots + t_m X_m})$$

$$= \exp\left\{ \sum_{i=1}^{m} t_i \mu_i + \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} t_i t_j \operatorname{Cov}(X_i, X_j) \right\}$$

which shows that joint dist^n of $X_1, \ldots, X_m$ MVN is completely determined by a knowledge of $E(X_i) = \mu_i$ and $\operatorname{Cov}(X_i, X_j)$, $i, j = 1, \ldots, m$.

Gaussian process:

Def^n A SP $\{X(t), t \geq 0\}$ is a Gaussian or a normal process if $X(t_1), \ldots, X(t_n)$ has MVN $\forall\ t_1, \ldots, t_n$

— × —

→ $\{B(t), t \geq 0\}$ BM process is a Gaussian process

Sol^n $B(t_1), B(t_2), \ldots, B(t_n)$ can be expressed as l.c. of indep Normal rv's $B(t_1), B(t_2) - B(t_1), \ldots, B(t_n) - B(t_{n-1})$

$$B(t_1) = B(t_1)$$

$$B(t_2) = B(t_1) + (B(t_2) - B(t_1))$$

indep

$$B(t_3) = B(t_1) + (B(t_2) - B(t_1)) + (B(t_3) - B(t_2))$$

indep

— x —

$W(t)$ Wiener / SBM process $\quad W(t) - N(0,t)$

$E(W(t)) = 0$

$\text{Cov}(W(s), W(t)) = E(W(s)W(t))$

$= E\left(W(s)\left(W(s) + W(t) - W(s)\right)\right)$ for $s < t$

$= E(W^2(s)) + E\left(\underbrace{W(s)}_{\sim}\underbrace{(W(t) - W(s))}_{\text{indep}}\right)$

$= s + E(W(s))\,E(W(t) - W(s))$

$= s$ for $s < t$

$= \min(s,t)$

$\longrightarrow$ A Gaussian sp $\{X(t)\}_{t \geq 0}$ is SBM iff

$m_X(t) := E(X_t) = 0$ and $r_X(s,t) = \text{Cov}(X(s), X(t))$

$= \min(s,t)$

, $s,t \geq 0$

Brownian Bridge:

For $s<t$

$[W(s)|W(t)=B] \sim N\left(\frac{s}{t}B, \frac{s(t-s)}{t}\right)$

$W(t)$ SBM

$[W(t), \ 0\le t\le 1 \mid W(1)=0]$

$W(t_1), \ldots, W(t_n)$ MVN, it follows that this conditional process, known as Brownian bridge (as it is tied down both at 0 and at 1) is a Gaussian process

$E(W(s)|W(1)=0) = 0$ for $s<1$

$s<t<1$

$\text{Cov}(W(s), W(t)|W(1)=0) = E(W(s)W(t)|W(1)=0)$

$= E\left(E(W(s)W(t)|W(t)) \mid W(1)=0\right)$

$= E\left(W(t)\, E(W(s)|W(t)) \mid W(1)=0\right)$

$= E\left(W(t)\, \frac{s}{t} W(t) \mid W(1)=0\right)$

$= \frac{s}{t} E\left(W^2(t) \mid W(1)=0\right)$

$= \frac{s}{t} \times \frac{t(1-t)}{1}$

$= s(1-t)$

$\therefore$ Brownian bridge can be defined as a Gaussian process with mean value 0 and Cov. fn $s(1-t)$ for $s<t$

Eg If $W(t)$ is BM, then $\{Z(t), 0 \le t \le 1\}$ is a Brownian bridge process with $Z(t) = W(t) - t\,W(1)$

Sol As $Z(t)$ is a Gaussian process, all we need to verify $E(Z(t)) = 0$ ✓, $Cov(Z(s), Z(t)) = s(1-t)$, $s < t$

$s < t$

$$Cov(Z(s), Z(t)) = Cov(W(s) - s\,W(1), W(t) - t\,W(1))$$
$$= Cov(W(s), W(t)) - t\,Cov(W(s), W(1)) - s\,Cov(W(1), W(t)) + st\,Cov(W(1), W(1))$$
$$= s - ts - st + st$$
$$= s(1-t)$$

— × —

Integrated BM :

$\{X(t), t \ge 0\}$ BM, then $Z(t) = \int_0^t X(s)\,ds$ is called integrated BM.

→ We want to model price of commodity throughout time

Let $Z(t)$ price at t

rather than assuming $Z(t)$ BM (or $\log Z(t)$ as BM) we might want to assume the rate of change of $Z(t)$ as BM

$$\therefore \frac{d}{dt} Z(t) = X(t)$$

$$\Rightarrow Z(t) = Z(0) + \int^t X(s)\,ds$$

Since BM is a Gaussian process

$\Rightarrow (Z(t), t \geq 0)$ is a Gaussian process

Assume $X(t) = W(t)$ SBM

$$E[Z(t)] = E\left(\int_0^t W(s)\,ds\right) = \int_0^t E(W(s))\,ds = 0$$

$s \leq t$, $\mathrm{Cov}(Z(s), Z(t)) = E(Z(s)Z(t))$

$$= E\left(\int_0^s W(y)\,dy \int_0^t W(u)\,du\right)$$

$$= E\left(\int_0^s \int_0^t W(y)W(u)\,dy\,du\right)$$

$$= \int_0^s \int_0^t E(W(y)W(u))\,dy\,du$$

$$= \int_0^s \int_0^t \min(y,u)\,dy\,du$$

$$= \int_0^s \left[\int_0^u y\,dy + \int_u^t u\,dy\right] du$$

$$= \int_0^s \left[\frac{u^2}{2} + u(t-u)\right] du$$

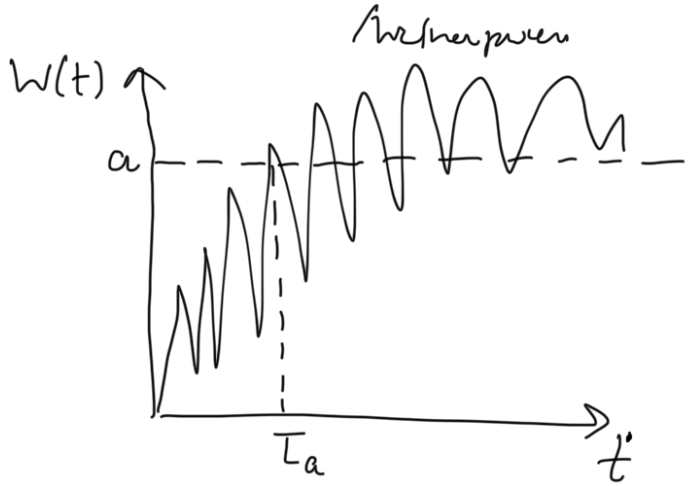
$$= \int_0^s \left[ut - \frac{u^2}{2}\right] du$$

$$= t\frac{s^2}{2} - \frac{s^3}{6} = s^2\left(\frac{t}{2} - \frac{s}{6}\right).$$

Hitting time and max. variable

~~Stopping time~~

τ_a : first time the SBM $W(t)$ hits a



For $a>0$

$$P(W(t)\geq a) = \underbrace{P(W(t)\geq a \mid \tau_a \leq t)}_{\frac{1}{2}} P(\tau_a \leq t) + \underbrace{P(W(t)\geq a \mid \tau_a > t)}_{0} P(\tau_a > t)$$

$$\Rightarrow P(\tau_a \leq t) = 2P(W(t)\geq a) \quad \text{——— (1)}$$

$$= \frac{2}{\sqrt{2\pi t}} \int_a^\infty e^{-x^2/2t}\, dx \leq \frac{2}{\sqrt{2\pi}} \int_{a/\sqrt{t}}^\infty e^{-y^2/2}\, dy$$

For $a<0$, the dist'y τ_a "by symmetry" the same as τ_{-a}

τ_a has CDF

$$\therefore P(\tau_a \leq t) = \frac{2}{\sqrt{2\pi}} \int_{|a|/\sqrt{t}}^\infty e^{-y^2/2}\, dy = 2\left(1-\Phi\left(\frac{|a|}{\sqrt{t}}\right)\right)$$

Φ → cdf of SN. variable

τ_a has pdf

$$f(t) = \frac{d}{dt} P(\tau_a \leq t) = \frac{|a|}{\sqrt{2\pi t^3}} e^{-a^2/2t}, \quad t>0$$

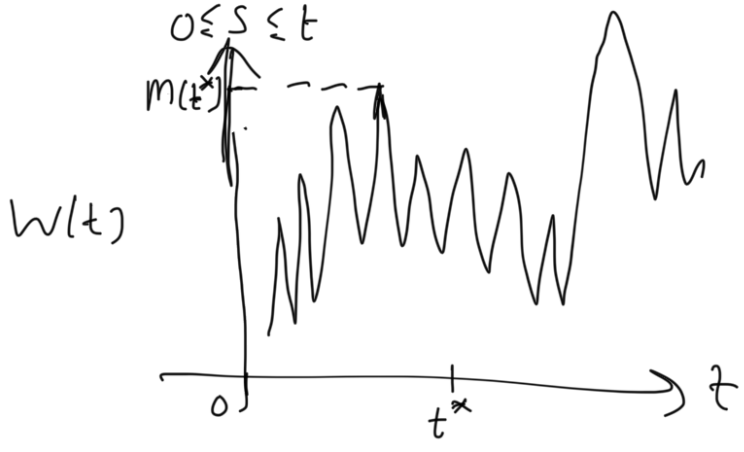
→ The hitting time τ_a is a.s. finite

So $P(\tau_a < \infty) = \lim_{t\to\infty} P(\tau_a \le t) = \lim_{t\to\infty} 2\left(1 - \Phi\left(\frac{|a|}{\sqrt{t}}\right)\right)$

$= 2\left(1 - \Phi(0)\right) = 2\left(1 - \frac{1}{2}\right) = 1$

→ Another variable of interest is the max. value of the process attains in $[0,t]$

$$M(t) = \max_{0 \le s \le t} W(s)$$



For $a > 0$

$$P(M(t) \ge a) = P\left(\max_{0 \le s \le t} W(s) \ge a\right)$$

$$= P(\tau_a \le t)$$

$$= 2\,P(W(t) \ge a) \quad \text{/ Using (1)}$$

$$= 2\left(1 - \Phi\left(\frac{a}{\sqrt{t}}\right)\right)$$

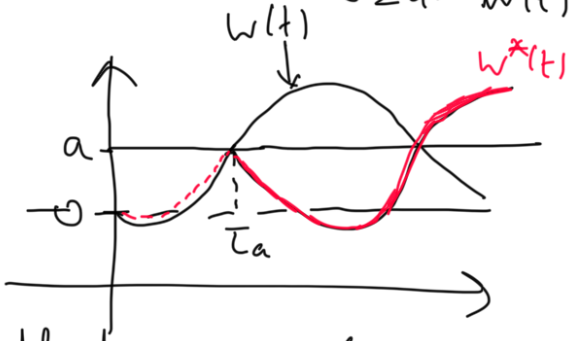
— x —

Thm: For any $a > 0$ and $y \ge 0$

$$P(W(t) \le a - y,\ M(t) > a) = P(W(t) > a + y)$$

$$= 1 - \Phi\left(\frac{a+y}{\sqrt{t}}\right)$$

Sol Define $W^*(t) = \begin{cases} W(t) & \text{if } t \le \tau_a \\ 2a - W(t) & \text{if } t > \tau_a \end{cases}$



∴ by reflection principle

$$P(W(t) \le a-y, M(t) > a) = P(W^*(t) > a+y)$$

$W^*(t)$ is Wiener process / SBM ?

↳ cont a.s.

$s < t \le \tau_a \quad W^*(t) - W^*(s) = W(t) - W(s)$

$t > s > \tau_a$

$$W^*(t) - W^*(s) = (2a - W(t)) - (2a - W(s))$$
$$= -(W(t) - W(s))$$
$$\sim N(0, t-s)$$

$s < \tau_a < t$

$$W^*(t) - W^*(s) = (2a - W(t)) - W(s)$$
$$= (a - W(s)) - (W(t) - a) \sim N(0, t-s)$$

After time τ_a, the process $W(t)$ is an indep SBM.

$$W(\tau_a) - W(s) \overset{\text{indep}}{-} W(t - \tau_a)$$

$$\overset{\sim}{=} W(T_a - s)$$

$$\sim$$

$$\lim_{t \uparrow T_a} W^*(t) = W(T_a) = a$$

$$\lim_{t \downarrow T_a} 2a - W(T_a) = a$$

Stochastic Calculus / Itô's integral:

$\{W(t), t \geq 0\}$ SBM / Wiener process

f having cont derivative on $[a,b]$

Sto. integral

$$\int_a^b f(t)\,dW(t) = \lim_{\substack{n\to\infty \\ \max(t_i - t_{i-1})\to 0}} \sum_{i=1}^{n} f(t_{i-1})\left(W(t_i) - W(t_{i-1})\right)$$

when $a = t_0 < t_1 < \cdots < t_n = b$ is a partition of $[a,b]$

$$\sum_{i=1}^{n} f(t_{i-1})\left(W(t_i) - W(t_{i-1})\right) = f(b)W(b) - f(a)W(a) - \sum_{i=1}^{n} W(t_i)\left(f(t_i) - f(t_{i-1})\right)$$

$$f(t_n)W(t_n) - f(t_0)W(t_0) - W(t_1)\left(f(t_1) - f(t_0)\right) - W(t_2)\left(f(t_2) - f(t_1)\right) - \cdots - W(t_n)\left(f(t_n) - f(t_{n-1})\right)$$

$$\boxed{\int_a^b f(t)\,dW(t) = f(b)W(b) - f(a)W(a) - \int_a^b W(t)\,df(t)}$$

$$E\left(\int_a^b f(t)\,dW(t)\right) = 0$$

$$V\left(\int_a^b f(t)\,dW(t)\right) = E\left(\int_a^b f(t)\,dW(t)\right)^2 = \int_a^b f^2(t)\,dt$$

(Itô's isometry)

Sol

$$V\left(\sum_{i=1}^{n} f(t_{i-1})\left(W(t_i) - W(t_{i-1})\right)\right) = \sum_{i=1}^{n} f^2(t_{i-1})\, V\left(W(t_i) - W(t_{i-1})\right)$$

$$V\left(W(t_i) - W(t_{i-1})\right) = (t_i - t_{i-1})$$

$$= \sum f^2(t_{i-1}) \cdots$$

$$\sum_{i=1}^{n} f(t_{i-1})\,(t_i - t_{i-1})$$

$f_t \equiv f(t)$ is non-random

$$\int_a^b f(t)\,dW(t) \sim N\left(0,\ \int_a^b f^2(t)\,dt\right)$$

Example ① Find the distr of $X := \int_0^t s\,dW_s$

$$X \sim N\left(0,\ \int_0^t s^2 ds\right) \equiv N\left(0,\ \frac{t^3}{3}\right)$$

② $$\int_0^t W_s\,dW_s \simeq \sum_j W_{t_j}\left(W_{t_{j+1}} - W_{t_j}\right)$$

$$= \frac{W_t^2}{2} - \frac{1}{2}\sum_j \left(W_{t_{j+1}} - W_{t_j}\right)^2 \longrightarrow \frac{W_t^2}{2} - \frac{t}{2}$$

$$a(b-a) = \frac{1}{2}(b^2 - a^2) - \frac{1}{2}(a-b)^2$$

$$\sum_{k=1}^{n} a_{k-1}(a_k - a_{k-1}) = \frac{1}{2}\sum_{k=1}^{n}\left(a_k^2 - a_{k-1}^2\right) - \frac{1}{2}\sum_{k=1}^{n}(a_k - a_{k-1})^2$$

$$= \frac{1}{2}a_n^2 - \frac{1}{2}\sum_{k=1}^{n}(a_k - a_{k-1})^2$$

$\sqrt{h}\,Z = W_{t+h} - W_t \sim N(0, h)$ where $Z \sim N(0,1)$

$$\sum_{j=1}^{n}\left(W_{tj/n} - W_{t(j-1)/n}\right)^2 \overset{d}{=} \sum_{j=1}^{n}\left(\sqrt{\frac{t}{n}}\,Z_j\right)^2$$

Z_j's are iid $N(0,1)$

$$= t\,\underbrace{\frac{1}{n}\sum_{j=1}^{n} Z_j^2}_{\to E(Z_j^2) = 1} \to t$$

—x—

$$\int_0^t W_s\,dW_s = \frac{W_t^2}{2} - \frac{t}{2}$$

$$W_t^2 = \int_0^t ds + 2\int_0^t W_s\,dW_s$$

Ex. $W_t^3 = 3\int_0^t W_s\,ds + 3\int_0^t W_s^2\,dW_s$

Itô formula 1

$(X_t)_{t\in[0,T]}$ is an Itô process (on filtered prob. space $(\Omega, \mathcal{F}, \mathbb{F}, P)$ with BM $(W_t)_{t\in[0,T]}$ on it) if

$$X_t = X_0 + \int_0^t a_s\,ds + \int_0^t b_s\,dW_s \quad , t\in[0,T] \qquad (1)$$

when $\to X_0$ is $\mathcal{F}_0$-mble

$\to (a_t = a_t(\omega))_{t\in[0,T]}$ is an $\mathbb{F}$-adapted process with mble trajectories s.t. $\int_0^T |a_s|\,ds < \infty$

$\to (b_t = b_t(\omega))_{t\in[0,T]}$ " " $\mathbb{F}$-adapted "

s.t. $\int_0^T E(b_s^2)\,ds < \infty$

—x—

(1) has on $[0,T]$ stochastic differential

$$dX_t = a_t\,dt + b_t\,dW_t \qquad (2)$$

Multiplication table

$$dt \cdot dt = (dt)^2 = 0$$

$$\sum_{j=} \frac{t}{n} \times \left(W_{\frac{j}{n}t} - W_{\frac{j-1}{n}}\right)$$

$$\sum (\Delta t_k)^2 \to 0$$

$dW_t\, dt = dt\, dW_t = 0$ — (1)

$dW_t\, dW_t = (dW_t)^2 = dt$

$\sum (\Delta t_k)(\Delta W_k) \to 0$

$\sum (\Delta W_k)^2 \to t$

—×—

$[X_t]$ Itô's process with sto. differential (2)

f cont., twice differentiable fun

$Y_t = f(X_t)$, $t \in [0,T]$

$$f(x) = f(a) + f_x(a)(x-a) + \frac{1}{2} f_{xx}(a)(x-a)^2$$

$$dY_t \equiv df(X_t) = f'(X_t)\, dX_t + \frac{1}{2} f''(X_t)\, \underbrace{(dX_t)^2}_{b_t^2\, dt}$$

Example ① W_t $\qquad f(x) = \frac{1}{2}x^2$

$d(f(W_t)) = d\left(\frac{1}{2} W_t^2\right)$ $\qquad f'(x) = x$, $f''(x) = 1$

$= f'(W_t)\, dW_t + \frac{1}{2} f''(W_t)\, (dW_t)^2$

$= W_t\, dW_t + \frac{1}{2}\, dt$

X_t Itô's process $\qquad dX_t = a_t\, dt + b_t\, dW_t$

$Y_t = f(t, X_t)$ is also an Itô's process

$$f(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$$

$$+ \frac{f_{xx}(a,b)}{2}(x-a)^2 + f_{xy}(a,b)(x-a)(y-b) + \frac{f_{yy}(a,b)}{2}(y-b)^2$$

$$dY_t = df(t,X_t) = \partial_t f(t,X_t)\,dt + \partial_x f(t,X_t)\,dX_t + \frac{1}{2} f_{xx}(t,X_t)\,\underbrace{(dX_t)^2}_{\| \; b_t^2\,dt}$$

Example Geometric B.M.

$$Z_t = e^{\mu t + \sigma W_t} \qquad ; \ \mu, \sigma \in \mathbb{R}$$

$$= f(t,W_t)$$

$$f(t,x) = e^{\mu t + \sigma x}$$

$$\partial_t f = \mu f(t,x) \ , \ \partial_x f = \sigma f(t,x) \ ,$$

$$\partial_{xx} f = \sigma^2 f(t,x)$$

$$d(Z_t) = d\big(f(t,W_t)\big)$$

$$= \partial_t f(t,W_t)dt + \partial_x f(t,W_t)\,dW_t + \frac{1}{2}\partial_{xx} f(t,W_t)\,(dW_t)^2$$

$$= \mu Z_t dt + \sigma Z_t dW_t + \frac{1}{2}\sigma^2 Z_t\,dt$$

$$= \left(\mu + \frac{\sigma^2}{2}\right) Z_t\,dt + \sigma Z_t\,dW_t$$

Product rule:

X_t, Y_t Itô process on common filtered prob. space

$$Z_t := X_t Y_t$$

$$dZ_t \equiv d(X_t Y_t) = X_t\, dY_t + Y_t\, dX_t + dX_t\, dY_t$$

Quotient rule $Z_t = X_t / Y_t$

Ex. $$dZ_t = d\left(\frac{X_t}{Y_t}\right) = \frac{dX_t}{Y_t} - \frac{X_t}{Y_t^2} dY_t + \frac{X_t}{Y_t^3}(dY_t)^2 - \frac{1}{Y_t^2} dX_t\, dY_t$$

Example: $$Z_t := W_t e^{W_t} = X_t Y_t$$

$$X_t = W_t, \quad Y_t = e^{W_t}$$

$$dY_t = d(e^{W_t}) = e^{W_t} dW_t + \frac{1}{2} e^{W_t} dt$$

$$dZ_t = X_t\, dY_t + Y_t\, dX_t + dX_t\, dY_t$$

$$= W_t\left(e^{W_t} dW_t + \frac{1}{2} e^{W_t} dt\right) + e^{W_t} dW_t + dW_t\left(e^{W_t} dW_t + \frac{1}{2} e^{W_t} dt\right)$$

$$= \left(1 + \frac{1}{2} W_t\right) e^{W_t} dt + (1 + W_t) e^{W_t} dW_t$$

Stochastic Differential Equations (SDE):

$$dX_t = a(t, X_t)\,dt + b(t, X_t)\,dW_t, \quad t \in [0,T], \qquad X_0 = x_0 \qquad \text{---(1)}$$

where $a(t,x)$, $b(t,x)$, $t \geq 0$, $x \in \mathbb{R}$ are non-random, $\{W_t\}$ SBM on $(\Omega, \mathcal{F}, \mathbb{F}, P)$.

Itô process $\{X_t\}_{t \in [0,T]}$ on $(\Omega, \mathcal{F}, \mathbb{F}, P)$ is said to be sol^n of SDE (1) if

$$X_t = x_0 + \int_0^t a(s, X_s)\,ds + \int_0^t b(s, X_s)\,dW_s, \quad t \in [0,T] \qquad \text{---(2)}$$

[It is important to point out the SDE's rarely admit sol^n of an analytic closed form.

Example (Ornstein–Uhlenbeck process) OU process

$$dX_t = -r X_t\,dt + \sigma\,dW_t, \qquad X_t|_{t=0} = X_0, \quad r, \sigma > 0 \qquad \text{---(3)}$$

$$e^{rt}\,dX_t = -r X_t e^{rt}\,dt + \sigma e^{rt}\,dW_t \qquad \text{---(4)}$$

$$f(t,x) = e^{rt}x, \quad \partial_t f = rf, \quad \partial_x f = e^{rt}, \quad \partial_{xx} f = 0$$

$$d(f(t, X_t)) = d(e^{rt} X_t)$$

$$= r e^{rt} X_t\,dt + e^{rt}\,dX_t \qquad \text{---(5)}$$

Using (5) in (4)

$$d(e^{rt}X_t) = \sigma e^{rt} dW_t$$

$$e^{rt}X_t - X_0 = \sigma \int_0^t e^{rs} dW_s$$

$$\Rightarrow X_t = e^{-rt}X_0 + \sigma \int_0^t e^{-r(t-s)} dW_s \text{ is a sol}^n \text{ of (3)}$$

Let $\theta_t = \int_0^t e^{-r(t-s)} dW_s$ is a Gaussian process

with $E(\theta_t) = 0$

$$E(\theta_t^2) = E\left(\int_0^t e^{-r(t-s)} dW_s \right)^2$$

Ito's isometry

$$= \int_0^t e^{-2r(t-s)} ds$$

$$= e^{-2rt} \left(\frac{e^{2rs}}{2r} \right)_{s=0}^{t}$$

$$= \frac{e^{-2rt}}{2r} \left(e^{2rt} - 1 \right) = \frac{1}{2r} (1 - e^{-2rt})$$

If $X_0 = x_0$ (6)

$$X_t \sim N(e^{-rt}x_0, \sigma_{r,t}^2), \text{ where } \sigma_{r,t}^2 = \frac{\sigma^2}{2r}(1 - e^{-2rt})$$

✡ X_t is a Gaussian process if X_0 is Gaussian

✡ $X_t \to N\left(0, \frac{\sigma^2}{2r}\right)$ $\quad t \to \infty$

— × —

Vasieck Interest rate model (VIR model)

OU process mean-reverting property

math model of interest rate dynamics

Spot interest rate r_t is assumed to follow SDE

$$dr_t = a(b - r_t)\,dt + \sigma\,dW_t, \quad t \geq 0$$

$$a, b, \sigma > 0, \quad r_0 \geq 0$$

let $X_t = r_t - b$, $dX_t = dr_t$

$$dX_t = -aX_t\,dt + \sigma\,dW_t$$

$r_t - b$ is OU process

$$r_t = X_t + b \sim N\left(b + e^{-at}(r_0 - b), \sigma^2_{a,t}\right)$$

| Using (6)

Cox - Ingersoll - Ross interest rate model

CIR model :

$$dr_t = a(b - r_t)\,dt + \sigma\sqrt{r_t}\,dW_t, \quad t \geq 0$$

$$a, b, \sigma, r_0 > 0$$

— × —

Geometric BM :

GBM is a sol^n of SDE with multiplicative noise

N_t : price of asset

r : interest rate

α : volatility

$$dN_t = r N_t\, dt + \alpha N_t\, dW_t \quad , \quad N_t|_{t=0} = N_0 \qquad (6)^*$$

$$\frac{dN_t}{N_t} = r\, dt + \alpha\, dW_t \qquad (7)$$

$d(\log N_t)$.

$f(t,x) = \log x$

$\partial_t f = 0, \ \partial_x f = \frac{1}{x}, \ \partial_{xx} f = -\frac{1}{x^2}$

$$= \frac{1}{N_t} dN_t - \frac{1}{2N_t^2}(dN_t)^2$$

$(dN_t)^2 = \alpha^2 N_t^2\, dt$

$= r\, dt + \alpha\, dW_t$ using (7)

$$\Rightarrow d(\log N_t) = \left(r - \frac{\alpha^2}{2}\right) dt + \alpha\, dW_t$$

$$\int_0^t d \log N_t = \left(r - \frac{\alpha^2}{2}\right) t + \alpha W_t$$

$$\log N_t - \log N_0 = \left(r - \frac{\alpha^2}{2}\right) t + \alpha W_t$$

$\Rightarrow N_t = N_0 \exp\left\{\left(r - \frac{\alpha^2}{2}\right) \ldots$

(9) Prove that

$$\int_0^\infty x^{m-1}\left(\frac{1}{x}-1\right)^{n-1}dx = \Gamma m\,\Gamma n$$

Markov inequality: $X \geq 0$

$$P(X \geq u) \leq \frac{E(X)}{u}$$

$$E(X) = \int_{-\infty}^{u} u f(u)\,du + \int_{u}^{\infty} u f(u)\,du$$

$$\geq \int_{u}^{\infty} u f(u)\,du$$

$$\geq u \int_{u}^{\infty} f(u)\,du$$

$$= u\, P(X \geq u)$$

Chebyshev: ξ mean μ, var σ^2

$$P(|\xi - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2}$$

for any $\epsilon > 0$

Choose $X = (\xi - \mu)^2$, $u = \epsilon^2$

$$\frac{E(\xi-\mu)^2}{\epsilon^2} \geq P((\xi-\mu)^2 \geq \epsilon^2)$$

$$\Rightarrow P(|\xi - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2} \quad \text{for any } \epsilon > 0$$

— x —

(1) $X \sim \exp$ with mean $E(X) = 1$

$u = 0.5$

Compare $P(X > u)$ with Markov inequality bound $\frac{E(X)}{u}$

$$P(X > u) = e^{-u}$$

$$P(X > \tfrac{1}{2}) = e^{-1/2} \qquad \frac{E(X)}{u} = \frac{1}{1/2} = 2$$

(2) $X \sim$ Bernoulli (p)

Compare $P(X \geq 1)$ with Markov inequality bound $E(X)$

$$P(X \geq 1) = P(X=1) = p \quad ; \underline{E(X)} = p$$

(3) Let $U_1, U_2, \ldots$ be indep rv's each uniformly distributed over the interval $[0,1]$. Show that $X_0 = 1$ and $X_n = 2^n U_1 \cdots U_n$ for $n = 1, 2, \ldots$ is a MG

Sol $X_{n+1} = 2 X_n U_{n+1}$

$$E(X_{n+1} \mid X_0, \ldots, X_n) = E(2 X_n U_{n+1} \mid X_0, \ldots, X_n)$$

$$= 2 X_n E(U_{n+1} \mid U_0, \ldots, U_n)$$

$$\overset{\text{indep of } U_0, \ldots, U_n}{=} 2 X_n E(U_{n+1})$$

$$= 2 X_n \cdot \frac{1}{2} \qquad E(U_{n+1}) \to \int_0^1 u \times 1 \, du = \frac{1}{2}$$

$$= X_n$$

$\{X_n\}$ is an MG

(4) Let $S_0 = 0$ for $n \geq 1$, $S_n = \epsilon_1 + \cdots + \epsilon_n$ be sum of n indep rvs, each expo. distributed with mean $E(\epsilon) = 1$. $X_n = 2^n e^{-S_n}$, $n \geq 0$ is MG?

Sol $X_{n+1} = 2 X_n e^{-\epsilon_{n+1}}$

$$E(X_{n+1} \mid X_0, \ldots, X_n) = E(2 X_n e^{-\epsilon_{n+1}} \mid X_0, \ldots, X_n)$$

$$= 2 X_n E(e^{-\epsilon_{n+1}} \mid \epsilon_0, \ldots, \epsilon_n)$$

$$\overset{\text{indep } \epsilon_1, \ldots, \epsilon_n}{=} 2 X_n E(e^{-\epsilon_{n+1}}) \qquad \left(\int_0^\infty e^{-u} \cdot 1 \, du = 1 \right.$$

$= 2X_n \cdot \frac{1}{2} = 1$

(5) Let $\xi_1, \xi_2, \ldots$ be indep Bernoulli rv's with parameter p, $0 < p < 1$, show that $X_0 = 1$ and $X_n = p^{-n} \xi_1 \cdots \xi_n$, $n = 1, 2, \ldots$ is MG

What is the limit X_n as $n \to \infty$.

Sol

$$X_{n+1} = p^{-1} \xi_{n+1} X_n$$

$$E(X_{n+1} \mid X_0, X_1, \ldots X_n) = E(p^{-1} \xi_{n+1} X_n \mid X_0 \ldots X_n)$$

$$= X_n E(p^{-1} \xi_{n+1} \mid X_0 \ldots X_n)$$

$$= X_n E(p^{-1} \xi_{n+1} \mid \xi_0, \xi_1 \ldots \xi_n)$$

by independence of ξ_i's $= X_n E(p^{-1} \xi_{n+1}) = X_n \times p^{-1} \times E(\xi_{n+1})$

$$= X_n$$

$$\xi_i = \begin{cases} 1 & \text{wp } p \\ 0 & \text{wp } 1-p \end{cases} \qquad \prod_{i=1}^{n} \xi_i = \begin{cases} 1 & \text{wp } p^n \\ 0 & \text{wp } 1-p^n \end{cases}$$

$$X_n = \begin{cases} \frac{1}{p^n} & \text{wp } p^n \\ 0 & \text{wp } 1-p^n \end{cases}$$

$n \to \infty$

$X_n \Rightarrow 0$ wp 1

(6) Consider a SP that evolves according to the following laws: If $X_n = 0$, then $X_{n+1} = 0$, whereas if $X_n > 0$, then

$$X_{n+1} = \begin{cases} X_n + 1 & \text{wp } \frac{1}{2} \\ X_n - 1 & \text{wp } \frac{1}{2} \end{cases}$$

sol X_n is MG.

$$E(X_{n+1} \mid X_0 = x_0, \cdots, X_n = x_n)$$

$$= (x_n + 1) \times \frac{1}{2} + (x_n - 1) \times \frac{1}{2}$$

$$= x_n$$

$$E(X_{n+1} \mid X_0, \cdots, X_n) = X_n$$

$\therefore X_n$ is MG.

Application in Finance I

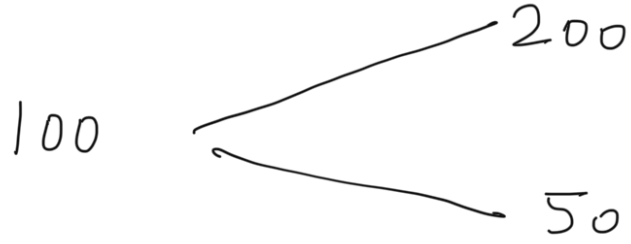
Pricing Stock Option:

amt x at time t in the future is not worth as much as given x immediately

x at time t is $x e^{-\alpha t}$

$\alpha \rightarrow$ discount factor

$e^{-\alpha t}$ discount fr



time 0 price time 1 price

Option Rs C $\rightarrow K = 150$ per share

Strike price

Suppose at time 0

buy x unit of stock

buy y " " options

Value of holding at time 1

$$\text{Value} = \begin{cases} 200x + 50y, & \text{if price is } 200 \\ 50x & \text{if price is } 50 \end{cases}$$

Let we choose y s.t.

$$200x + 50y = 50x$$

$$\Rightarrow y = -3x$$

If $y = -3x$, the value of holding at time 1 is $50x$

Original cost $= 100x - 3xc$

x unit, $y = -3x$ units of option

$c \to$ unit cost of option

$$\text{gain} = 50x - (100x - 3xc)$$

$$= x(3c - 50)$$

$$= \begin{cases} 0 & \text{if } 3c = 50 \\ +ve & \text{if } 3c > 50 \\ -ve & \text{if } 3c < 50 \end{cases}$$

eg (1) if $c = 20$ (unit cost per option)

$x = 1$, $y = -3$

purchase 1 unit of stock and simultaneously sell 3 unit of option initially cost is $= 100 - 60 = 40$

which holding at time 1 is 50

guranteed profit $= 50 - 40 = Rs\,10$ is attained

(2) If $c = 15$

$x = -1$, $y = 3$

initial gain $= 100 - 3 \times 15 = 55$

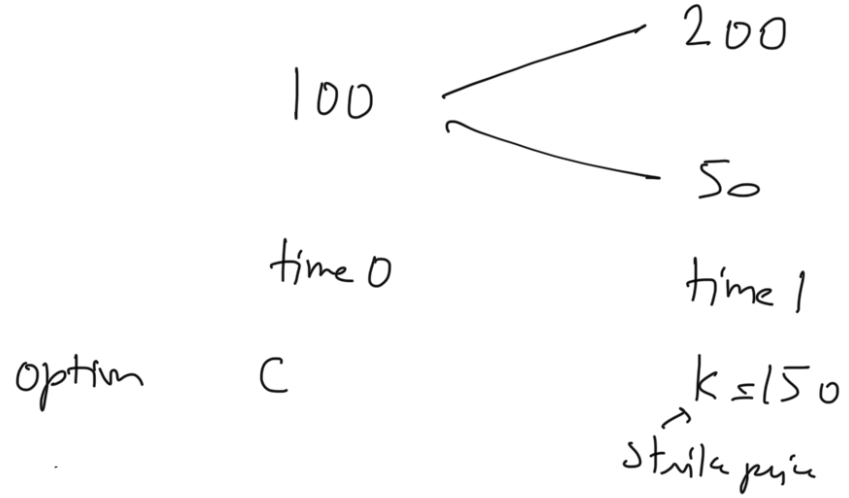
Value of holding at time 1 $= -50$

guranted profit $= 55 - 50 = Rs\ 5$

—x—

A sure win betting scheme is called an arbitrage (risk free profit). Thus there is this example option cost c that does not result in an arbitrage is $c = 50/3$.

—x—



$(p, 1-p)$

Wagers $\longrightarrow$ Stock, option

$E(\text{return Wager}) = 0$

return from purchasing 1 unit of stock $= \begin{cases} 200 - 100 = 100, & \text{prob } p \\ 50 - 100 = -50, & \text{,, ,, } 1-p \end{cases}$

$E(\text{return stock}) = 0$

$100p - 50(1-p) = 0$

$100 = 200p + 50(1-p)$

$$\Rightarrow p = \frac{1}{3}$$

return from purchasing one share of option $= \begin{cases} 50 - c & \text{if price is } 200 \\ -c & \text{if price is } 50 \end{cases}$

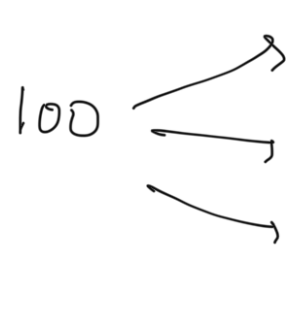
$E(\text{return option}) = 0$

$$(50 - c) \times \frac{1}{3} - c \times \frac{2}{3} = 0$$

$$c = (200 - 150) \times \frac{1}{3}$$

$$\Rightarrow c = \frac{50}{3}$$

— x —



$0 < p_1 < 1, 0 < p_2 < 1$

time 0 time 1

C K = 150

$$\begin{cases} 100 = 300 p_1 + 200 p_2 + 50 (1 - p_1 - p_2) \\ C = 150 p_1 + 50 p_2 \end{cases}$$

$0 < p_1 < 1, \; 0 < p_2 < 1$

Arbitrage thm: expt whose set of possible outcomes

$S = \{1, 2, \ldots, m\}$

n wagers

amt x is bet on wager i, then return $x\, r_i(j)$ is earned if the outcome of expt is $j \in S$

$r_i(\cdot) \to$ return fn for unit bet on wager i

Betting scheme $\underset{\sim}{x} = (x_1, \ldots, x_n)$

out of expt is j, then

return fn $\underset{\sim}{x} = \sum_{i=1}^{n} x_i\, r_i(j)$

$\exists\ \underset{\sim}{p} = (p_1, \ldots, p_m)$ in set of possible outcomes of under which each of wager has expected return 0 or else there is a betting scheme that guarantees a positive win.

Thm (Arbitrage thm) Exactly one of the following is true

Either

(i) $\exists$ a $\underset{\sim}{p} = (p_1, \ldots, p_m)$ for which

$$\sum_{j=1}^{m} p_j (r_i(j)) = 0 \quad \forall\ i = 1, \ldots, n$$

or

(ii) $\exists$ a betting scheme $\underset{\sim}{x} = (x_1, \ldots, x_n)$ for which

$$\sum_{i=1}^{n} x_i\, r_i(j) > 0 \quad \forall\ j = 1, \ldots, m$$

— x —

Forwards are contracts that give the market participant the right to buy or sell an underlying or financial asset at a time T in the future or in a future time interval (T, T') at a fixed price k.

- long position : entering into a contract to buy
- short position " " " sale contract

Forwards and futures involve hedging for their settlement, i.e, the market participant has both right and the obligation to buy or sell.

While futures are traded on financial markets, forwards are based on an individual agreement by the participant without market intervention

$x^+ = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$

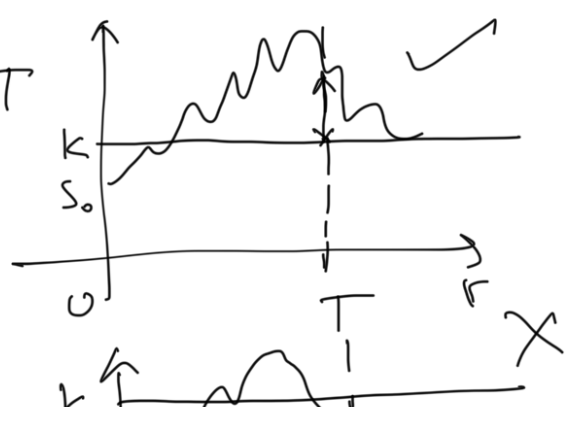
$= \max(x, 0)$

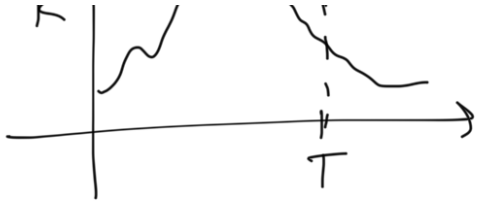
Call option

European call

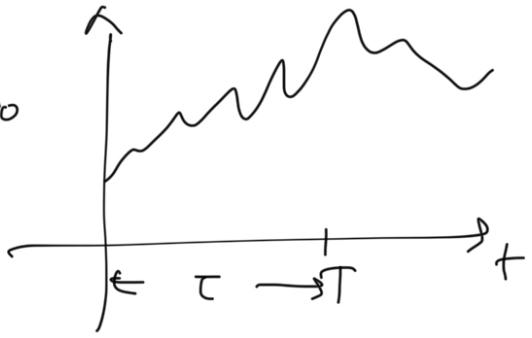
Value of European call at time T

$X = (S_T - k)^+$





American call
gives the buyer right to buy a security at any time in time $[0,T]$ at a price k



$\tau \in [0,T]$

price of American call $X = (S_\tau - k)^+$

Put option right to sell

$$X = (k - S_T)^+$$

→ Mathematically a derivative security (or contingent claim) with maturity (expiry) date T in our market is a

$$X = X(\omega) = g(S_T(\omega)) \geq 0$$

of the underlying asset price S_T at time T.
The financial interpretation of X is that the contract will pay its owner the amount X at time T.

S_t ; B_t, $r \geq 0$
$B_t = (1+r)^t B_0$

— X —

$p = 0.4 \rightarrow 130 = S_1^a$

$S_0 = 100$ $\xrightarrow{1-p=0.6}$ S_1^b

$T=0$ $T=1$

C_0 $k = 110$ European call at $T=1$

$r = 0.05$ $X = (S_1 - k)^+$

There is a risk-free investment at interest rate $r = 0.05$.

→ Classical way $t=1$ $E_p(X) = 20 \times 0.4 = 8$

∴ Value of call at $t=0$ is

$$C_0 = \frac{1}{1+r} E_p(X) = \frac{8}{1.05} = 7.62$$

→ Black - Scholes $C_0^{BS} = 9.52$

Determine portfolio which at $T=1$ has the same payoff as the call.

$\underset{\sim}{\Pi} = (\Delta, B)$

portfolio ↓ ↓ bond

stock share

$\Delta = 0.4$, $B = -30.98$

price of portfolio at $t=0$

$$V_0(\Pi) = \Delta \times 100 + B = 40 - 30.98 = 9.52$$

at $t=1$

$$V_1(\Pi) = \Delta S_1 + B(1+r)$$

$$= \begin{cases} 0.4 \times 130 - 30.48 \times 1.05 & , S_1 = 130 \\ \end{cases}$$

$$= \begin{cases} 0.4 \times 80 - 30.48 \times 1.05 , & S_1 = 80 \\ 20 \quad \text{if} \quad S_1 = 130 \\ 0 \quad \text{if} \quad S_1 = 80 \end{cases}$$

$$= X \quad = (S_1 - k)^+$$

$\therefore$ price of duplicating portfolio $\doteq$ price of call

$$\Pi = (\Delta, B) = (0.4, -30.48)$$

$\therefore$ Fair price of the call at $t=0$ is price of portfolio

$$C_0^{BS} = 100 \times 0.4 - 30.48$$

$$= 40 - 30.48 = 9.52$$

If one were to call the call differently, then a risk-free profit (arbitrage) would be possible. This price is based on no-arbitrage principle.

—x—

Example : ① $d\cos(W_t)$

$Y_t = f(t, X_t)$

$dY_t = \partial_t f(t,X_t)\,dt + \partial_x f(t,X_t)\,dX_t + \frac{1}{2}\partial_{xx} f(t,X_t)(dX_t)^2$

$f(t,x) = \cos x$

$\partial_t f = 0$, $\partial_x f = -\sin x$, $\partial_{xx} f = -\cos x$

$$d\cos W_t = 0 - \sin(W_t)\,dW_t - \frac{1}{2}\cos(W_t)\,dt$$

$$(dW_t)^2 = dt$$

— x —

② The price S_t of a risky asset evolves according to SDE

$$dS_t = 0{,}2\,S_t\,dt + S_t\,dW_t \quad , t \geq 0 , \quad S_0 = 5$$

(i) It is suspected that SDE has sol^n of the form $S_t = c\,e^{at+bW_t}$, a, b, c consts

Use Itô's formula to verify this suspicion and find the consts a, b, c

Sol $S_t = c\,e^{at+bW_t}$

$f(t,x) = c\,e^{at+bx}$

$\partial_t f = ac\,e^{at+bx}$

$\partial_x f = bc\,e^{at+bx}$, $\partial_{xx} f = b^2 c\,e^{at+bx}$

$dS_t = d\,f(t, W_t)$

$= ac\,e^{at+bW_t}\,dt + bc\,e^{at+bW_t}\,dW_t$

$$+ \frac{1}{2} b^2 c\, e^{at+bW_t} \underbrace{(dW_t)^2}_{\sim\, dt}$$

$$= a S_t\, dt + b S_t\, dW_t + \frac{1}{2} b^2 S_t\, dt$$

$$= \left(a + \frac{b^2}{2}\right) S_t\, dt + b S_t\, dW_t$$

$$W_0 = 0$$

$$S_0 = 5$$

$$c e^0 = 5 \Rightarrow c = 5$$

$$a + \frac{b^2}{2} = 0.2 \Rightarrow a = -0.3$$

$$b = 1 \qquad b = 1$$

$$c = 5$$

$$S_t = 5 e^{-0.3t + W_t}$$

(ii) Show that the process $X_t = \frac{1}{S_t}$ satisfies the SDE

$$dX_t = 0.8 X_t\, dt - X_t\, dW_t, \quad t \geq 0, \; X_0 = 0.2$$

$$X_t = \frac{1}{S_t} = \frac{1}{5 e^{-0.3t + W_t}} = 0.2\, e^{0.3t - W_t}$$

$$= f(t, W_t)$$

$$f(t, x) = 0.2\, e^{0.3t - x}$$

$$\partial_t f = 0.3 \times 0.2\, e^{0.3t - x} = 0.3 f$$

$$\partial_x f = -0.2\, e^{0.3t - x} = -f$$

$\partial_{xx} f = 0.2 e^{0.3t - x} = f$

$dX_t = df(t, X_t)$

$= 0.3 f(t, W_t)\, dt - f(t, W_t)\, dW_t + \frac{1}{2} f(t, W_t)(dW_t)^2$

$= 0.3 X_t dt - X_t dW_t + \frac{1}{2} X_t dt$

$= 0.8 X_t dt - X_t dW_t$

$\longrightarrow\!\times\!\longrightarrow$

Example The "stochastic volatility" Heston model assumes that the "variance process" $(V_t)_{t \geq 0}$ follows the SDE

$$dV_t = (1 - V_t)\, dt + 2\sqrt{V_t}\, dW_t \,, \quad t \geq 0, \; V_0 = 1$$

(i) Derive an SDE for the "volatility process" $Z_t = \sqrt{V_t}$, $t \geq 0$ and find the initial condition for the SDE (i.e. the value of Z_0)

Sol

$$dZ_t = d\sqrt{V_t} = df(t, V_t)$$

$f(t, x) = \sqrt{x}$

$\partial_t f = 0$, $\partial_x f = \frac{1}{2} x^{-1/2}$, $\partial_{xx} f = -\frac{1}{4} x^{-3/2}$, $x \geq 0$

$dZ_t = df(t, V_t)$

$= \frac{1}{2} V_t^{-1/2} dV_t - \frac{1}{8} V_t^{-3/2} (dV_t)^2$

$$= \frac{1}{2} V_t^{-1/2} \left((1 - V_t) dt + 2\sqrt{V_t}\, dW_t \right) - \frac{1}{8} V_t^{-3/2} \times 4 V_t\, dt$$

$$= dW_t + \frac{1}{2} V_t^{-1/2} dt - \frac{1}{2} V_t^{1/2} dt - \frac{1}{2} V_t^{-1/2} dt$$

$$= dW_t - \frac{1}{2} Z_t\, dt$$

$$dZ_t = dW_t - \frac{1}{2} Z_t\, dt$$

$$Z_0 = \sqrt{V_0} = 1$$

(ii) Show that the S.P.

$$Z_t := e^{-t/2} \left(1 + \int_0^t e^{s/2}\, dW_s \right)$$

satisfies the SDE and the initial condition you derived in part (i)

Set $Z_t = X_t Y_t$,

where $X_t = e^{-t/2}$

$$Y_t = 1 + \int_0^t e^{s/2}\, dW_s$$

$$dX_t = -\frac{1}{2} e^{-t/2} dt = -\frac{X_t}{2} dt$$

$$dY_t = e^{t/2} dW_t$$

$$dZ_t = d(X_t Y_t)$$

$$= X_t\, dY_t + Y_t\, dX_t + dX_t\, dY_t$$

$$= e^{-t/2} e^{t/2} dW_t - \frac{1}{2} e^{-t/2} \left(1 + \int_0^t e^{s/2} dW_s \right) dt - \frac{1}{2} e^{-t/2} e^{t/2} dW_t\, dt$$

$$= e^{t/2} e^{-t/2} dW_t - Y_t \frac{X_t}{2} dt - \frac{X_t}{2} e^{t/2} \underbrace{dt\, dW_t}_{=0}$$

$$= dW_t - \frac{1}{2} Z_t\, dt$$

Example (1) Let $\tau = \min\{t>0 : W_t = \pm\sqrt{a+bt}\}$ be the first time the BM crosses one of the two parabolic boundaries $\pm\sqrt{a+bt}$, $t\geq 0$, where $a>0$ and $b\in(0,1)$ are some constants. Compute $E(\tau)$

Sol $W_0=0$, $E(W_0^2)=0$ | $Y_t = W_t^2 - t\}_{t\geq 0}$ MG (see class notes)

$$0 = E(Y_0) = E(Y_\tau) \quad \text{/Using MG stopping thm}$$

$$= E(W_\tau^2 - \tau)$$

$$\Rightarrow E(W_\tau^2) - E(\tau) = 0$$

$$\Rightarrow E(a+b\tau) - E(\tau) = 0$$

$$\Rightarrow a + b\,E(\tau) - E(\tau) = 0$$

$$\Rightarrow E(\tau) = \frac{a}{1-b}$$

W_t: $E(W_t)=0$, $V(W_t) = E(W_t^2) = t$

(2) Denote by $\tau = \min\{t>0 : W_t \leq 2t-4\}$ the first time the BM process crosses the boundary $V_t := 2t-4$, $t\geq 0$.

$E(\tau)$, $\mathrm{Var}(\tau)$, $E(W_\tau)$, $E(W_\tau^2)$, Laplace transform $l_\tau(s) = E(e^{-s\tau})$

Sol W_t MG, $W_t^2 - t$ MG

$$0 = E(W_0) = E(W_\tau) = E(2\tau - 4)$$

$$\Rightarrow 2\,E(\tau) - 4 = 0 \Rightarrow E(\tau) = 2$$

$$\{Y_t := W_t^2 - t\}_{t \geq 0}$$

$$0 = E(Y_0) = E(Y_\tau) = E(W_\tau^2 - \tau)$$

$$= E(W_\tau^2) - E(\tau)$$

$$= E((2\tau - 4)^2) - E(\tau)$$

$$= E(4\tau^2 + 16 - 16\tau) - E(\tau)$$

$$= 4E(\tau^2) + 16 - 32 - 2$$

W_t mg $\qquad E(\tau^2) = 4.5$

$$0 = E(W_0) = E(W_\tau)$$

$W_t^2 - t$ mg

$$0 = E(W_0^2 - 0) = E(W_\tau^2 - \tau)$$

$$\Rightarrow E(W_\tau^2) = E(\tau) = 2$$

$\rightarrow \{Z_t := e^{uW_t - u^2 t/2}\}_{t \geq 0}$ mg

$$1 = E(Z_0) = E(Z_\tau) = E(e^{uW_\tau - u^2\tau/2})$$

$$= E(e^{u(2\tau - 4) - u^2\tau/2})$$

$$= E(e^{-4u - (\frac{u^2}{2} - 2u)\tau})$$

$$= e^{-4u} E(e^{-s\tau}), \text{ where } s = \frac{u^2}{2} - 2u$$

$$\Rightarrow E(e^{-s\tau}) = e^{4u}$$

$\swarrow$ solve

$$u = 2 \pm \sqrt{2(s+2)}$$

$$\ell_\tau(s) = E(e^{-s\tau}) = \exp\{8 - 4\sqrt{2(s+2)}\}$$

Example: Denote by $\tau := \min\{t>0 : W_t = a$ or $W_t = b\}$ the first time the SBM / Wiener process takes one of the values a or b $(a<0<b)$

(i) find the dist of W_τ

(ii) $E(\tau)$

Sol $W_\tau \in \{a, b\}$

$$P(W_\tau = a) = 1 - P(W_\tau = b)$$

W_t mg

$$0 = E(W_0) = E(W_\tau) = a\,P(W_\tau = a) + b\,(1 - P(W_\tau = a))$$

$$\Rightarrow \quad P(W_\tau = a) = \frac{b}{b-a}$$

$$P(W_\tau = b) = 1 - \frac{b}{b-a} = \frac{-a}{b-a}$$

$Y_t := W_t^2 - t$ mg

$$0 = E(Y_0) = E(Y_\tau) = E(W_\tau^2 - \tau)$$

$$= E(W_\tau^2) - E(\tau)$$

$$E(\tau) = E(W_\tau^2)$$

$$= a^2 \times \frac{b}{b-a} - b^2 \frac{a}{b-a}$$

$$= \frac{ab(a-b)}{} = -ab.$$

Application in Finance I

Option Pricing and Arbitrage Theory

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October 15, 2026

Introduction to Pricing Stock Options

- An amount v at time t in the future is not worth as much as given v immediately.
- **Present Value Calculation:**

$$ve^{-\alpha t}$$

where α is the discount factor, and $e^{-\alpha t}$ is the discount function.

Binomial Pricing Model

Consider a stock price movement from time 0 to time 1:

Option Specifications

- **Option Price:** Rs C
- **Strike Price (k):** 150 per share

Portfolio Construction

Suppose at time 0, a portfolio is formed by:

- Buying x units of stock
- Buying y units of options

Value of the holding at time 1:

$$\text{Value} = \begin{cases} 200x + 50y, & \text{if price is 200} \\ 50x, & \text{if price is 50} \end{cases}$$

Risk-Free Setup & Arbitrage Examples

To eliminate risk, choose y such that:

$$200x + 50y = 50x \implies y = -3x$$

If $y = -3x$, the time 1 portfolio value is certain to be $50x$.

Example 1: If $C = 20$

- Set $x = 1, y = -3$ (Buy 1 stock, sell 3 options).
- **Initial Cost:** $100 - 3(20) = 40$.
- **Time 1 Value:** 50.
- **Guaranteed Profit:** $50 - 40 =$ Rs 10 (Arbitrage!).

Risk-Neutral Probabilities

A sure-win betting scheme is called an **arbitrage** (risk-free profit). To avoid arbitrage, the option price must balance expected returns to 0.

Using probabilities p (up) and $1-p$ (down):

$$E(\text{Return Stock}) = 0 \implies 100p - 50(1-p) = 0 \implies p = \frac{1}{3}$$

Evaluating the option expected return under this probability:

$$(50 - C) \times \frac{1}{3} - C \times \frac{2}{3} = 0 \implies C = \frac{50}{3}$$

The General Arbitrage Theorem

Let an experiment have outcomes $S = \{1, 2, \ldots, m\}$ and n wagers. $r_i(j)$ represents the return per unit bet on wager i given outcome j.

Arbitrage Theorem

Exactly one of the following conditions is true:

1. **No Arbitrage:** $\exists$ a probability vector $\mathbf{p}$ such that every wager has an expected return of 0:

$$\sum_{j=1}^{m} p_j r_i(j) = 0 \quad \forall i$$

2. **Arbitrage Opportunity:** $\exists$ a betting scheme $\mathbf{x}$ that guarantees a strictly positive win:

$$\sum_{i=1}^{n} x_i r_i(j) > 0 \quad \forall j$$

Financial Derivatives Definitions

Forwards/Futures Contracts giving the right to buy or sell an asset at a future time T at a fixed price k. Long position = buy; short position = sell.

European Call Right to buy at maturity T. Payoff: $X = (S_T - k)^+$.

American Call Right to buy at any time $\tau \in [0, T]$. Payoff: $X = (S_\tau - k)^+$.

Put Option Right to sell at maturity T. Payoff: $X = (k - S_T)^+$.

Replicating Portfolios (Black-Scholes Principle)

Given $S_0 = 100$, tracking to either $S_1^u = 130$ or $S_1^d = 80$ at $T = 1$.
European call strike $k = 110$, risk-free rate $r = 0.05$.

- **Replication Strategy:** Form a portfolio $\Pi = (\Delta, B)$ with Δ stock shares and B bonds.
- Set $\Delta = 0.4$ and $B = -30.98$.
- **Initial Value:** $V_0 = 0.4(100) - 30.98 = 9.52$.

Time 1 Payoff Replication

$$V_1 = \begin{cases} 0.4(130) - 30.98(1.05) = 20, & \text{if } S_1 = 130 \\ 0.4(80) - 30.98(1.05) = 0, & \text{if } S_1 = 80 \end{cases}$$

This perfectly mirrors the option. Thus, the unique no-arbitrage price is **9.52**.

General discrete model for securities

$S_n = (S_n^0, S_n^1, \dots, S_n^d)$ → d development of d equity securities.

let S_n^i : price of ith stock at time n

$S_n^0 = (1+r)^n$

→ A portfolio describes the proportions of different securities in the portfolio

→ A trading strategy $\underset{\sim}{\Phi} = (\underset{\sim}{\Phi}_n) = (\underset{\sim}{\Phi}_n^0, \dots, \underset{\sim}{\Phi}_n^d)$ describes the development of the portfolio

$\underset{\sim}{\Phi}_n^i$ ~ # shares of security number i at time n.

Value of portfolio at time n is

$$V_n(\underset{\sim}{\Phi}) = \underset{\sim}{\Phi}_n \cdot \underset{\sim}{S}_n = \sum_{i=0}^{d} \underset{\sim}{\Phi}_n^i S_n^i$$

→ $\underset{\sim}{\Phi}$ is called self financing if

$$\underset{\sim}{\Phi}_n \cdot \underset{\sim}{S}_n = \underset{\sim}{\Phi}_{n+1} \cdot \underset{\sim}{S}_n$$

Changes to the portfolio result only from regrouping. No additional capital is required for the change. No withdrawal of value take place.

→ increase in value is

$$\Delta V_n(\phi) = V_{n+1}(\phi) - V_n(\phi)$$

$$= \phi_{n+1} \cdot S_{n+1} - \phi_n \cdot S_n$$

$$= \phi_{n+1} \cdot S_{n+1} - \phi_{n+1} \cdot S_n \quad \left| \text{ if } \phi \text{ is self financing} \right.$$

$$= \phi_{n+1} \cdot \Delta S_n$$

→ The no-arbitrage principle implies: There is no-arbitrage strategy ϕ, i.e., there is no strategy ϕ s.t.

$V_0(\phi) = 0$, $V_n(\phi) \geq 0$ and $P(V_n(\phi) > 0) > 0$,

or equivalent to it

$V_0(\phi) \leq 0$, $V_n(\phi) \geq 0$ and $P(V_n(\phi) > 0) > 0$.

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Binomial Market

$$t = 0, 1, \ldots, T$$

- bond (or bank account) yielding a riskless rate r of interest in each time period

$$B_t = (1+r)^t, \quad t = 0, 1, \ldots, T$$

- a risky asset (stock) at time t, S_t

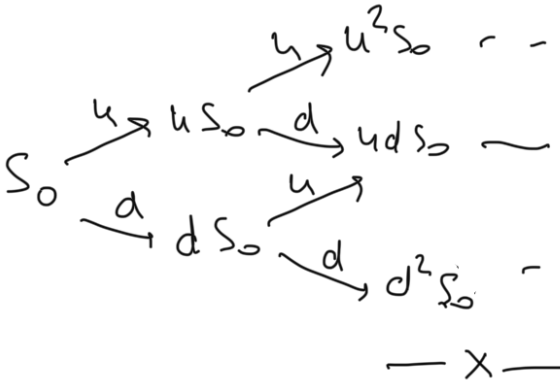
$$\rightarrow S_t = u S_{t-1}$$

$S_{t-1} \nearrow S_t = d S_{t-1}$, $t = 1, 2, \dots, T$

$d < 1 + r < u$

$S_t = S_t(\omega) = V_1 \cdots V_t S_0$ | $V_j = d$ or u

price process is adapted to filtration $\{\mathcal{F}_t\}$



— x —

Single-period Binomial market :

trading strategy (Δ, b) at $t = 0$

Δ: # of shares, b: # of bonds

time t value of portfolio

$$V_t = V_t(\omega) = \begin{cases} \Delta S_0 + b B_0 = \Delta S_0 + b & \text{at } t = 0 \\ \Delta S_1 + b B_1 = \Delta S_1 + b(1+r) & \text{at } t = 1 \end{cases}$$

The portfolio will be a hedge given that

$V_1(\omega) \geq X(\omega)$, $\omega \in \Omega = \{u, d\}$

$T = 1$ $X(\omega) = X_u$

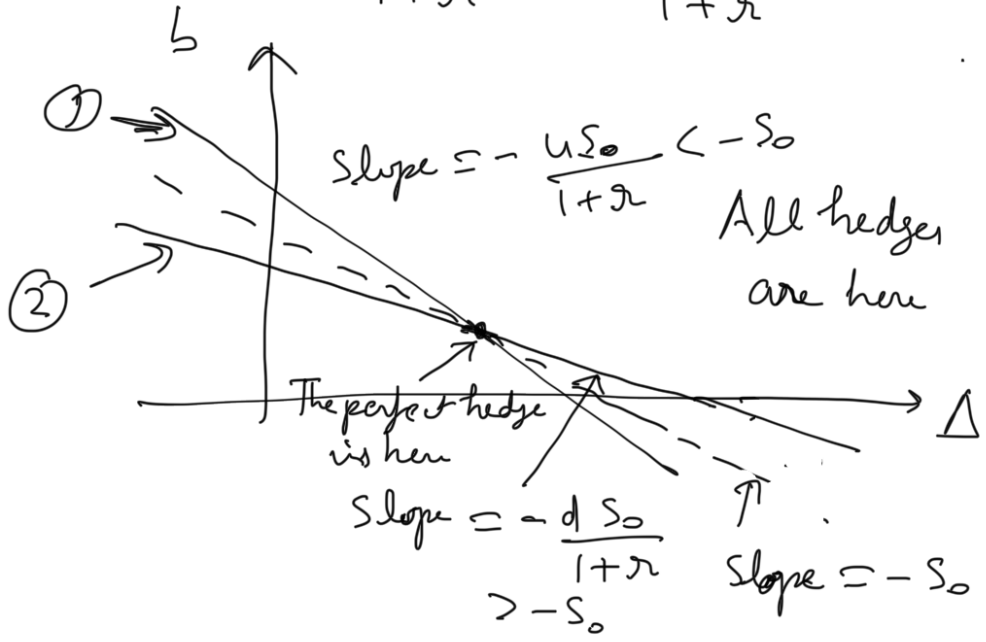
$X(\omega) = X_d$

hedge conditions $\begin{cases} \Delta u S_0 + b(1+r) \geq X_u \end{cases}$

$$\begin{cases} \Delta d S_0 + b(1+r) \geq X_d \end{cases}$$

$$\Rightarrow \quad b \geq -\frac{uS_0}{1+r}\Delta + \frac{X_u}{1+r} \qquad \text{—— (1)}$$

$$b \geq \frac{-dS_0}{1+r}\Delta + \frac{X_d}{1+r} \qquad \text{—— (2)}$$



$$\Delta u S_0 + b(1+r) = X_u$$

$$\Delta d S_0 + b(1+r) = X_d$$

$$\Rightarrow \Delta = \frac{X_u - X_d}{(u-d)S_0} = \frac{g(uS_0) - g(dS_0)}{uS_0 - dS_0} \qquad \text{—— (3)}$$

$$b = \frac{X_u - \Delta u S_0}{1+r} = \frac{uX_d - dX_u}{(1+r)(u-d)} \qquad \text{—— (4)}$$

(3), (4) is a perfect hedge for X and its time $t=0$ value is

$$V_0 = \Delta S_0 + b$$

$$= \frac{X_u - X_d}{u-d} + \frac{uX_d - dX_u}{(1+r)(u-d)}$$

$$= \frac{\cdots}{u-d} + \frac{u X_d - d X_u}{(1+r)(u-d)}$$

$$= \frac{1}{1+r}\left[\underbrace{\frac{1+r-d}{u-d}}_{=: p^*} X_u + \underbrace{\frac{u-(1+r)}{u-d}}_{= 1-p^*} X_d \right]$$

$$= \frac{1}{1+r} E^*(X) = E^*\left(\frac{X}{1+r}\right) = X^*$$

where E^* denote expectation under the prob. p^* on $\Omega = \{u, d\}$ given by

$$p^*(\{u\}) = 1 - p^*(\{d\}) \quad , \text{where } p^* = \frac{1+r-d}{u-d} \in (0,1)$$

owing to condition, $d < 1+r < u$.

We maintain that thus calculated X^* is the fair price of the claim X time $t=0$.

Indeed, if the claim was sold at a different price $X_0 \neq X^*$ at that time, that would have created an arbitrage opportunity.

$$E_{p^*}(S_1 | S_0) = p^* u S_0 + (1-p^*) d S_0$$

$$= \frac{1+r-d}{u-d} u S_0 + \frac{u-(1+r)}{u-d} d S_0$$

$$= (1+r) S_0$$

$$\Rightarrow \quad E_{p^*}\left(\frac{S_1}{1+r} \mid S_0\right) = S_0$$

$d < 1+r < u$

(NA) no-arbitrage we has

$$S_0 \in \left(\frac{S_1(d)}{1+r}, \frac{S_1(u)}{1+r}\right)$$

This means that the pt. S_0 can be represented as convex combination of the interval end pts:

i.e., (for $p^* \in (0,1)$)

$$S_0 = p^* \frac{S_1(u)}{1+r} + (1-p^*)\frac{S_1(d)}{1+r} = E^*\left(\frac{S_1}{1+r}\right)$$

Example (Pricing a European Call)

Let $S_0 = 1$, $k = 1$, $r = 0.25$, $u = 1.75$, $d = 0.5$

$$p^* = \frac{(1+r)-d}{u-d} = \frac{1.25-0.5}{1.75-0.5} = 0.6$$

$$X^* = \frac{1}{1+r}\left[p^* X_u + (1-p^*) X_d\right]$$

$$= \frac{1}{1.25}\left[0.6 \times (1.75-1)^+ + 0.4 \times (0.5-1)^+\right)$$

$$= \frac{1}{1.25}\left[0.6 \times 0.75\right] = 0.36$$

replicating portfolio

$$\Delta = \frac{X_u - X_d}{}$$

$$\Delta = \frac{x_u - x_d}{(u-d)S_0} = \frac{0.75 - 0}{1.75 - 0.5} = 0.6$$

$$b = \frac{u x_d - d x_u}{(u-d)(1+r)} = -\frac{0.5 \times 0.75}{1.25 \times 1.25} = -0.24$$

$$(\Delta, b) = (0.6, -0.24)$$

$t=1$

$$V_1(u) = \Delta u S_0 + b(1+r) = 0.6 \times 1.75 + (-0.24) \times 1.25$$
$$= 0.75$$

$$V_1(d) = \Delta d S_0 + b(1+r) = 0.6 \times 0.5 + (-0.24) \times 1.25$$
$$= 0$$

$\therefore$ price of call is

$$V_0 = \Delta S_0 + b = 0.6 \times 1 + (-0.24) = 0.36 = x^* \quad (= C)$$

$\longrightarrow$ price a European put

$$P = \frac{1}{1+r}\left[p^*(uS_0 - k)^- + (1-p^*)(dS_0 - k)^-\right]$$

$$= \frac{1}{1.25}\left[0.6 \times (1.75-1)^- + 0.4\,(0.5-1)^-\right]$$

$$= \frac{1}{1.25}(0.4 \times 0.5) = 0.16$$

"put call parity"

C, P, k in NA

portfolio
- one share of the stock
- one put on one share of stock

- a short position in one call on one share of the stock

$t=0$ $\quad V_0 = S_0 + P - C$ ✓

$t=1$ $\quad V_1 = S_1 + (S_1-k)^- - (S_1-k)^+$

$= S_1 - \left((S_1-k)^+ - (S_1-k)^-\right)$

$= S_1 - (S_1-k) = k$ ✓

due to NA $\quad E\left(\frac{V_1}{1+r}\right) = V_0$

$$\frac{k}{1+r} = V_0$$

$$\therefore \quad S_0 + P - C = \frac{k}{1+r}$$

$$\therefore P = \frac{k}{1+r} - S_0 + C = \frac{1}{1.25} - 1 + 0.36$$

$$= 0.16$$

—x—

General Discrete Models and Binomial Markets

Portfolio Strategies, Arbitrage, and Option Pricing

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General Discrete Model for Securities

Consider a market developing over discrete time steps with d equity securities:

$$S_n = (S_n^0, S_n^1, \ldots, S_n^d) \quad \text{— development of } d \text{ equity securities}$$

Market Components

- **Stock Prices:** Let S_n^i be the price of the i-th stock at time n.
- **Riskless Asset (Bond):** $S_n^0 = (1+r)^n$, where r is the risk-free interest rate per period.

Definition (Portfolio & Trading Strategy)

A **portfolio** describes the proportions of different securities held.
A **trading strategy** $\Phi = (\Phi_n) = (\Phi_n^0, \Phi_n^1, \ldots, \Phi_n^d)$ describes the evolution of the portfolio, where Φ_n^i is the number of shares of security i held at time n.

Value of Portfolio and Self-Financing Property

Value of Portfolio

The total value of the portfolio Φ at time n is defined by the inner product:

$$V_n(\Phi) = \Phi_n \cdot S_n = \sum_{i=0}^{d} \Phi_n^i S_n^i$$

Definition (Self-Financing Strategy)

A trading strategy Φ is called **self-financing** if:

$$\Phi_n \cdot S_n = \Phi_{n+1} \cdot S_n$$

This implies that changes to the portfolio result **only from regrouping**. No additional capital is required, and no withdrawal of value takes place.

Value Increments and the No-Arbitrage Principle

For a self-financing strategy, the increase in portfolio value is given by:

$$\Delta V_n(\Phi) = V_{n+1}(\Phi) - V_n(\Phi) = \Phi_{n+1} \cdot S_{n+1} - \Phi_{n+1} \cdot S_n = \Phi_{n+1} \cdot \Delta S_n$$

The No-Arbitrage Principle

There is no self-financing strategy Φ that creates a riskless profit from nothing. That is, there is no strategy Φ satisfying:

$$V_0(\Phi) = 0, \quad V_n(\Phi) \geq 0, \quad \text{and } P(V_n(\Phi) > 0) > 0$$

Or equivalently:

$$V_0(\Phi) \leq 0, \quad V_n(\Phi) \geq 0, \quad \text{and } P(V_n(\Phi) > 0) > 0$$

The Binomial Market Model

Let time $t = 0, 1, \ldots, T$. The market consists of two assets:

1. Riskless Bond / Bank Account

Yields a riskless rate of interest r in each time period:

$$B_t = (1 + r)^t, \quad t = 0, 1, \ldots, T$$

2. Risky Asset (Stock)

At time t, the stock price evolves as:

$$S_t = \begin{cases} uS_{t-1} & \text{(Up move)} \\ dS_{t-1} & \text{(Down move)} \end{cases}$$

with the condition: $d < 1 + r < u$

The price process is **adapted to the filtration** $(\mathcal{F}_t)$, meaning $S_t(\omega) = v_1 v_2 \cdots v_t S_0$ where $v_j \in \{u, d\}$.

Single-Period Binomial Market ($T = 1$)

Let a trading strategy at $t = 0$ be denoted by (Δ, b), where:

- Δ = number of shares of stock
- b = number of bonds

Portfolio Value Equations

$$V_t = V_t(\omega) = \begin{cases} \Delta S_0 + bB_0 = \Delta S_0 + b & \text{at } t = 0 \\ \Delta S_1 + bB_1 = \Delta S_1 + b(1 + r) & \text{at } t = 1 \end{cases}$$

To **hedge** a contingent claim $X(\omega)$ with $X(u) = X_u$ and $X(d) = X_d$, the portfolio must satisfy the hedging conditions:

$$\Delta u S_0 + b(1 + r) \geq X_u$$
$$\Delta d S_0 + b(1 + r) \geq X_d$$

Finding the Perfect Hedge

To find the **perfect hedge**, both inequalities must hold as exact equalities:

$$\Delta u S_0 + b(1+r) = X_u$$
$$\Delta d S_0 + b(1+r) = X_d$$

Solving this linear system yields the replicating portfolio parameters:

Perfect Hedge Solution

$$\Delta = \frac{X_u - X_d}{(u-d)S_0} = \frac{g(uS_0) - g(dS_0)}{uS_0 - dS_0}$$
$$b = \frac{X_u - \Delta u S_0}{1+r} = \frac{uX_d - dX_u}{(1+r)(u-d)}$$

Fair Price and Risk-Neutral Measure

Substituting Δ and b back into the initial portfolio value $V_0 = \Delta S_0 + b$:

$$V_0 = \frac{1}{1+r}\left[\left(\frac{1+r-d}{u-d}\right)X_u + \left(\frac{u-(1+r)}{u-d}\right)X_d\right]$$

Risk-Neutral Probabilities

Define $p^* = \frac{1+r-d}{u-d}$. Since $d < 1+r < u$, it follows that $p^* \in (0,1)$. Thus:

$$V_0 = \frac{1}{1+r}\mathbb{E}^*[X] = \mathbb{E}^*\left[\frac{X}{1+r}\right] = X^*$$

where $\mathbb{E}^*$ is the expectation under the risk-neutral probability $P^*(\{u\}) = p^*, P^*(\{d\}) = 1-p^*$.

Fair Pricing

X^* is the unique **fair price** of the claim X at time $t = 0$. Any other price $X_0 \neq X^*$ creates an arbitrage opportunity. Notice that $\mathbb{E}^*\left[\frac{S_1}{1+r}\middle| S_0\right] = S_0$.

Example: Pricing a European Call Option

Given parameters: $S_0 = 1$, $K = 1$, $r = 0.25$, $u = 1.75$, $d = 0.5$.

- **1. Calculate the Risk-Neutral Probability p^*:**

$$p^* = \frac{(1+r)-d}{u-d} = \frac{1.25-0.5}{1.75-0.5} = \frac{0.75}{1.25} = 0.6 \implies (1-p^*) = 0.4$$

- **2. Calculate the Option Payoffs at $t = 1$:**

$$X_u = (1.75-1)^+ = 0.75, \quad X_d = (0.5-1)^+ = 0$$

- **3. Find the Fair Price X^*:**

$$X^* = \frac{1}{1.25}\left[0.6 \times 0.75 + 0.4 \times 0\right] = \frac{0.45}{1.25} = 0.36$$

Example: Replicating Portfolio & Put Option Pricing

Replicating Portfolio for the Call

$$\Delta = \frac{0.75 - 0}{(1.75 - 0.5) \times 1} = 0.6, \quad b = \frac{0.5 \times 0.75}{(1.25) \times (1.25)} = -0.24$$

At $t = 0$, $V_0 = 0.6(1) - 0.24 = 0.36 = X^*$.

Pricing a European Put Option (P)

Using the risk-neutral formula for a Put option where payoff is $(K - S_1)^+$:

$$P = \frac{1}{1+r}\left[p^*(uS_0 - K)^- + (1-p^*)(dS_0 - K)^-\right]$$

$$P = \frac{1}{1.25}\left[0.6 \times 0 + 0.4 \times (1 - 0.5)\right] = \frac{1}{1.25}[0.4 \times 0.5] = 0.16$$

Put-Call Parity

Consider a portfolio set up at $t = 0$: **Long 1 share of stock, Long 1 Put, Short 1 Call**.

$$V_0 = S_0 + P - C$$

At time $t = 1$, the value becomes deterministic regardless of stock movement:

$$\begin{aligned} V_1 &= S_1 + (K - S_1)^+ - (S_1 - K)^+ \\ &= S_1 - \left[(S_1 - K)^+ - (K - S_1)^+\right] \\ &= S_1 - (S_1 - K) = K \end{aligned}$$

By the No-Arbitrage principle, the initial value must equal the discounted final sure payoff:

$$V_0 = \frac{K}{1+r} \implies S_0 + P - C = \frac{K}{1+r}$$

Verification with Example Data

$$P = \frac{K}{1+r} - S_0 + C = \frac{1}{1.25} - 1 + 0.36 = 0.8 - 1 + 0.36 = 0.16$$

Matches our directly calculated Put price perfectly!

Finite Single period market

Single-period Binomial market, the NA (no-arbitrage) condition has the form

$$S_0 \in \left(\frac{S_1(d)}{1+r}, \frac{S_1(u)}{1+r} \right) \quad \exists \text{ EMM}$$

(eqv. martingale measure)

p^* on $\Omega = \{u, d\}$ s.t.

$$S_0 = p^* \frac{S_1(d)}{1+r} + (1-p^*) \frac{S_1(d)}{1+r} = E^* \left(\frac{S_1}{1+r} \right)$$

$t = 0, t = 1$

$t = 1 \quad \Omega = \{\omega_1, \dots, \omega_N\}, \quad N < \infty$

$(n+1)$ traded assets in the market

- n stocks $S_t = \{S_{t,1}, \dots, S_{t,n}\}$
- 1 bond/bank account $B_t = (1+r)^t, \quad t = 0, 1.$

→ A set $D \subset \mathbb{R}^m$ is called convex if for any $\underset{\sim}{x}_1, \underset{\sim}{x}_2 \in D$, all the points on the st. line segment connecting the x_i's also lie in D:

$$\beta \underset{\sim}{x}_1 + (1-\beta) \underset{\sim}{x}_2 \in D, \quad \beta \in (0,1)$$

Convex hull of a set $B \subset \mathbb{R}^m$, $\text{conv}(B)$, is the smallest convex set D s.t. $B \subset D$.

For a finite $B := \{y_1, \dots, y_N\} \subset \mathbb{R}^m$ one has

$$D := \mathrm{Conv}(B) = \left\{ \underset{\sim}{x} \in \mathbb{R}^m : \underset{\sim}{x} = \sum_{j=1}^{N} \alpha_j \underset{\sim}{y}_j, \alpha_j \geq 0, \sum_{j=1}^{N} \alpha_j = 1 \right\}$$

— x —

→ Denote by D_0 the relative interior of convex hull D:

$$D_0 = \left\{ \underset{\sim}{x} \in \mathbb{R}^m : \underset{\sim}{x} = \sum_{j=1}^{N} \alpha_j \underset{\sim}{y}_j, \alpha_j > 0, \sum_{j=1}^{N} \alpha_j = 1 \right\}$$

— x —

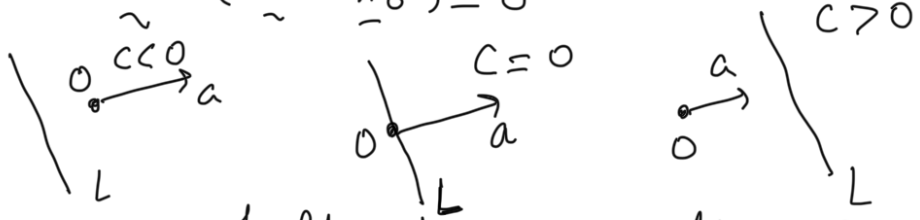
→ By a hyperplane in $\mathbb{R}^m$ one means a linear manifold of dim $m-1$. Analytically, a hyperplane can be specified by one linear equation

$$L := \{ \underset{\sim}{x} = (x_1, \dots, x_m) \in \mathbb{R}^m : \underset{\sim}{a} \cdot \underset{\sim}{x} = c \},$$

where $\underset{\sim}{a} \cdot \underset{\sim}{x} = \sum_{j=1}^{m} x_j a_j$, $c \in \mathbb{R}$, $\underset{\sim}{a} = (a_1, \dots, a_m) \neq 0$ is fixed

→ A vector $\underset{\sim}{a}$ is orthogonal to L: for any $\underset{\sim}{x}_1, \underset{\sim}{x}_2 \in L$ one has $\underset{\sim}{a} \cdot (\underset{\sim}{x}_1 - \underset{\sim}{x}_2) = 0$, and that $\underset{\sim}{x}_0 \in L$, the eqn for L can be written as

$$\underset{\sim}{a} \cdot (\underset{\sim}{x} - \underset{\sim}{x}_0) = 0$$



→ The open half-space on the side of L at which $\underset{\sim}{a}$ is pointing can be given as

$$H_L(\underset{\sim}{a}) := \{ \underset{\sim}{x} \in \mathbb{R}^m : \underset{\sim}{a} \cdot (\underset{\sim}{x} - \underset{\sim}{x}_0) > 0 \}$$

Thm1 seperation thm Assume that $D \in \mathbb{R}^m$ is convex set, D_0 its relative interior. Then for any pt $\underset{\sim}{x}_0 \notin D_0$, $\exists$ a hyperplane L with a normal vector $\underset{\sim}{a}$ s.t $\underset{\sim}{x}_0 \in L$ and $D_0 \subset H_L(\underset{\sim}{a})$.

In other words $\exists$ a vector $\underset{\sim}{a} \in \mathbb{R}^m$ s.t.

$$\underset{\sim}{a}(\underset{\sim}{x} - \underset{\sim}{x}_0) > 0, \quad \forall \ \underset{\sim}{x} \in D_0$$

— × —

Thm2 A finite single-period financial market is arbitrage-free iff $\exists$ a prob. P^* on $(\Omega, 2^\Omega)$ s.t $P^*(\{\omega\}) > 0$ for every $\omega \in \Omega$ and

$$E^*\left(\frac{\underset{\sim}{S}_1}{1+r}\right) = \underset{\sim}{S}_0, \text{ i.e.}$$

the process $\left\{\underset{\sim}{S}_t (1+r)^{-t}\right\}_{t=0,1}$ is a (vector-valued) MG (martingale) under P^*.

Sol $n = 2$ $\underset{\sim}{S}_t = (S_{t,1}, S_{t,2})$, $t = 0, 1$

two risky ones

set $D := \text{Conv}\left\{\frac{\underset{\sim}{S}_1(\omega_1)}{1+r}, \cdots, \frac{\underset{\sim}{S}_1(\omega_N)}{1+r}\right\}$

Two possibilities:

either (I) $\underset{\sim}{S}_0 \in D_0$, or (II) $\underset{\sim}{S}_0 \notin D_0$

We will prove that NA $\Leftrightarrow \exists$ EMM in three steps

Step1 shows $\exists$ EMM $\Leftrightarrow$ (I), Step2 proves the $\exists$ EMM $\Rightarrow$ NA, Step3 : NA $\Rightarrow$ (I)

Step1 from def^n of D_0

$$S_0 = \sum_{k=1}^{N} p_k^* \frac{S_1(\omega_k)}{1+r} = E^*\left(\frac{S_1}{1+r}\right)$$

thus $\exists$ EMM $\Leftrightarrow$ (I)

Step2 Pf by contradiction: Suppose that $\exists$ EMM holds, but NA does not

, ie, arbitrage opportunity $(\underset{\sim}{\Delta}, b)$ s.t. $V_t = \underset{\sim}{\Delta} \cdot \underset{\sim}{S}_t + B_t$

$$V_0 = \underset{\sim}{\Delta} \cdot \underset{\sim}{S}_0 + b = 0 \quad (\text{so that } b = -\underset{\sim}{\Delta} \cdot \underset{\sim}{S}_0)$$

but $V_1(\omega) \equiv \underset{\sim}{\Delta} \cdot S_1(\omega) + b(1+r)$

$$= \underset{\sim}{\Delta} \cdot \underset{\sim}{S}_1(\omega) - \underset{\sim}{\Delta} \cdot \underset{\sim}{S}_0 (1+r)$$

$$\begin{cases} \geq 0 & , \forall \omega \\ > 0 & \text{for at least one } \omega \end{cases} \quad \text{—— } \otimes$$

However, this means that

$$\underset{\sim}{\Delta} \cdot \underset{\sim}{S}_0 \overset{\text{Step1}}{=} \underset{\sim}{\Delta} \cdot E^* \frac{\underset{\sim}{S}_1}{1+r} = E^* \frac{\underset{\sim}{\Delta} \cdot \underset{\sim}{S}_1}{1+r}$$

$$= \sum_{k=1}^{N} p_k^* \frac{\underset{\sim}{\Delta} \cdot \underset{\sim}{S}_1(\omega_k)}{1+r}$$

$$\overset{\text{why } \otimes}{>} \sum_{k=1}^{N} p_k^* \, \underset{\sim}{\Delta} \cdot \underset{\sim}{S}_0 = \underset{\sim}{\Delta} \cdot \underset{\sim}{S}_0$$

(Contradiction)

$\therefore$ $\exists$ EMM $\Rightarrow$ NA

Step3 This also we will prove by contradiction

Suppose NA holds ... but (I) ...

suppose NA holds, but (I) does not,

latter means that (II) holds and then by the seperation thm $\exists$ a s.t. line L seperating S_0 from D_0, Denoting by $\underset{\sim}{\Delta} := (\Delta_1, \Delta_2)$ a normal to L vector "pointing" in the direction of D_0 we have

$$\underset{\sim}{\Delta} \cdot (\underset{\sim}{x} - \underset{\sim}{S_0}) > 0 \quad , x \in D_0$$

$$\underset{\sim}{\Delta} \cdot \left(\frac{S_1(\omega)}{1+r} - S_0 \right) \begin{cases} \geq 0 & \forall \omega \\ > 0 & \text{for atleast one } \omega \end{cases}$$

But this is arbitrage opportunity

Contradiction

NA $\Rightarrow$ (I)

— x —

Thm 3 (The arbitrage pricing thm) : In a single-period arbitrage-free market, the discounted value process $\{ V_t (1+r)^{-t} \}_{t=0,1}$ of any trading strategy $(\underset{\sim}{\Delta}, b)$ in an MG under any EMM p^*,

$$E^* \left(\frac{V_1}{1+r} \right) = V_0 . \qquad \text{——— (1)}$$

In particular, if X is an attainable claim in the market, then its time $t=0$ value

is given by

$$X^* := E^*\left(\frac{X}{1+r}\right) \quad -(2)$$

where the RHS does not depend on the choice of the EMM p^*

(A claim is considered attainable if $\exists$ a self-financing strategy that matches the claim capital at the maturity:
The strategy is called a hedging strategy)

Set $(\underset{\sim}{\Delta}, b)$

$$E^*\left(\frac{V_1}{1+r}\right) = E^*\left(\frac{\underset{\sim}{\Delta}\cdot S_1 + b(1+r)}{1+r}\right)$$

$$= E^*\left(\frac{\underset{\sim}{\Delta}\cdot \underset{\sim}{S_1}}{1+r}\right) + b$$

$$= \underset{\sim}{\Delta}\, E^*\left(\frac{\underset{\sim}{S_1}}{1+r}\right) + b$$

$$= \underset{\sim}{\Delta}\cdot \underset{\sim}{S_0} + b \quad \Big| \text{ using thm 2}$$

$$= \underset{\sim}{V_0} \quad -(3)$$

$\therefore$ (1) holds

Now suppose X is an attainable claim, i.e., for some portfolio (Δ, b)

portfolio (Δ, b) one has

$$V_1(\omega) = X(\omega), \quad \omega \in \Omega \quad \text{—— (4)}$$

(1) & (2) implies that

$$V_0 = E^*\left(\frac{V_1}{1+r}\right) = E^*\left(\frac{X}{1+r}\right) = X^* \quad \text{— (5)}$$

which proves (2).

→ The value of X^* does not depend on the choice of EMM p^* follows from (5), as the LHS in that relation is one and the same for any EMM.

→ Thm 3 provides one with the arbitrage-free price for any attainable claim. It turns out that, in the general case, not all claims are attainable in a financial market.

A market with the property that any claim is attainable in it is said to be complete.

Single-period NA binomial market is complete.

Thm 4 (The completeness thm): An arbitrage-free market is complete s/f $\exists$ (there exists) a unique EMM.

Sol

→ Observation 1 using thm 2 $\exists\, p^*$ s.t. $E^*\left(\frac{S_1}{1+r}\right) = S_0$

$$p_1^* S_1^1(\omega_1) + \cdots + p_N^* S_1^1(\omega_N) = (1+r) S_0^1$$

$$\vdots$$

$$p_1^* S_1^n(\omega_1) + \cdots + p_N^* S_1^n(\omega_N) = (1+r) S_0^n$$

$$p_1^* + p_2^* + \cdots + p_N^* = 1$$

So if EMM is unique, the above system has unique sol'n

$n+1 = N$

system matrix $A = \begin{pmatrix} S_1^1(\omega_1) & S_1^1(\omega_2) & \cdots & S_1^1(\omega_N) \\ S_1^2(\omega_1) & S_1^2(\omega_2) & \cdots & S_1^2(\omega_N) \\ \vdots & & & \\ S_1^n(\omega_1) & S_1^n(\omega_2) & \cdots & S_1^n(\omega_N) \end{pmatrix}$ is non-singular

$\det A \neq 0$ and hence A^{-1}

→ Observation 2 We observe claim $X = X(\omega)$ can be replicated when $\exists$ portfolio $(\underset{\sim}{\Delta}, b)$, $\underset{\sim}{\Delta} = (\Delta_1, \ldots, \Delta_n)$

$$\left.\begin{aligned} \Delta_1 S_1^1(\omega_1) + \cdots + \Delta_n S_1^n(\omega_1) + b(1+r) &= X(\omega_1) \\ &\vdots \\ \Delta_1 S_1^1(\omega_N) + \cdots + \Delta_n S_1^n(\omega_N) + b(1+r) &= X(\omega_N) \end{aligned}\right\} \text{✡}_0$$

system of N linear equations for $(n+1)$ unknowns $\Delta_1 \to \Delta_n$, $b(1+r)$ and system matrix is A^T

($\Leftarrow$) Suppose $\exists$ a unique EMM from obs. 1, $n+1 = N$ and A is non-singular, and hence so is A^T

since $\det A = \det A^T$

$\therefore$ Φ_0 has unique sol^n which means that, for any claim X, there is unique replicating strategy in the market, which proves that the market is complete.

($\Rightarrow$) Thm2 for an NA market, $\exists$ an EMM

We want to show if market is complet then EMM is unique

assume $\exists$ two EMM $\underset{\sim}{p}^*$, $\underset{\sim}{\tilde{p}}^*$. Due to completeness, by thm3, we have that for any claim X

$$E^*\left(\frac{X}{1+r}\right) = \tilde{E}^*\left(\frac{X}{1+r}\right)$$

$$\Leftrightarrow \sum_{k=1}^{N} p_k^* X(\omega_k) = \sum_{k=1}^{N} \tilde{p}_k^* X(\omega_k) \quad \checkmark \quad (6)$$

Take $X_i(\omega_k) := \delta_{ik} = \begin{cases} 1 & \text{if } k=i \\ 0 & \text{o.w} \end{cases}$, $i = 1, \dots, N$

$$p_i^* = \sum_{k=1}^{N} p_k^* \delta_{ik} = \sum_{k=1}^{N} p_k^* X_i(\omega_k) = \sum_{k=1}^{N} \tilde{p}_k^* X_i(\omega_k)$$

(why (6))

$$= \sum_{k=1}^{N} \tilde{p}_k^* \delta_{ik} = \tilde{p}_k^*$$

$\therefore$ $\underset{\sim}{p}^* = \underset{\sim}{\tilde{p}}^*$, meaning that EMM is unique.

RK: Market should be rich enough for us to be able to replicate an arbitrary claim.

Finite Single-Period Financial Markets

No-Arbitrage, Valuation, and Completeness

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Finite Single-Period Market Model

Market Structure ($t = 0, t = 1$)

Consider a financial market with a finite sample space $\Omega = \{\omega_1, \ldots, \omega_N\}$ where $N < \infty$. There are $(n+1)$ traded assets:

- n **Risky Stocks:** $\boldsymbol{S}_t = [S_{t,1}, \ldots, S_{t,n}]^T$ for $t = 0, 1$.
- **1 Risk-Free Asset (Bond/Bank Account):** $B_t = (1+r)^t$ for $t = 0, 1$, with interest rate $r > 0$.

The Binomial Case ($n = 1, N = 2$)

In a single-period binomial market where $\Omega = \{d, u\}$, the **No-Arbitrage (NA)** condition states:

$$S_0 \in \left(\frac{S_1(d)}{1+r}, \frac{S_1(u)}{1+r} \right)$$

This implies the existence of an **Equivalent Martingale Measure (EMM)** p^* such that:

$$S_0 = p^* \frac{S_1(u)}{1+r} + (1-p^*) \frac{S_1(d)}{1+r} = \mathbb{E}^* \left[\frac{S_1}{1+r} \right]$$

Geometric Foundations: Convexity and Hyperplanes

Definition (Convex Set & Convex Hull)

A set $D \subset \mathbb{R}^m$ is **convex** if for any $\boldsymbol{x}_1, \boldsymbol{x}_2 \in D$, the line segment connecting them also lies in D:

$$\beta \boldsymbol{x}_1 + (1-\beta)\boldsymbol{x}_2 \in D, \quad \forall \beta \in (0,1)$$

The **convex hull** $\text{conv}(B)$ of a finite set $B = \{\boldsymbol{y}_1, \ldots, \boldsymbol{y}_N\} \subset \mathbb{R}^m$ is defined as:

$$D = \text{conv}(B) = \left\{ \boldsymbol{x} \in \mathbb{R}^m : \boldsymbol{x} = \sum_{j=1}^{N} \alpha_j \boldsymbol{y}_j,\ \alpha_j \geq 0,\ \sum_{j=1}^{N} \alpha_j = 1 \right\}$$

Definition (Relative Interior)

The **relative interior** D_0 of the convex hull D is given by requiring strictly positive weights:

$$D_0 = \left\{ \boldsymbol{x} \in \mathbb{R}^m : \boldsymbol{x} = \sum_{j=1}^{N} \alpha_j \boldsymbol{y}_j,\ \alpha_j > 0,\ \sum_{j=1}^{N} \alpha_j = 1 \right\}$$

Hyperplanes and the Separation Theorem

Hyperplanes in $\mathbb{R}^m$

A hyperplane L is an $(m-1)$-dimensional linear manifold specified by a normal vector $\boldsymbol{a} = (a_1, \ldots, a_m) \neq \mathbf{0}$ and $c \in \mathbb{R}$:

$$L = \{\boldsymbol{x} \in \mathbb{R}^m : \boldsymbol{a} \cdot \boldsymbol{x} = c\}$$

An open half-space on the side of L towards which $\boldsymbol{a}$ points is defined as:

$$H_L(\boldsymbol{a}) := \{\boldsymbol{x} \in \mathbb{R}^m : \boldsymbol{a} \cdot (\boldsymbol{x} - \boldsymbol{x}_0) > 0\}, \quad \text{where } \boldsymbol{x}_0 \in L$$

Theorem (Separation Theorem)

Assume $D \subset \mathbb{R}^m$ is a convex set and D_0 is its relative interior. For any point $\boldsymbol{x}_0 \notin D_0$, there exists a hyperplane L with normal vector $\boldsymbol{a} \in \mathbb{R}^m$ such that $\boldsymbol{x}_0 \in L$ and $D_0 \subset H_L(\boldsymbol{a})$. That is:

$$\exists \boldsymbol{a} \in \mathbb{R}^m \quad s.t. \quad \boldsymbol{a} \cdot (\boldsymbol{x} - \boldsymbol{x}_0) > 0, \quad \forall \boldsymbol{x} \in D_0$$

The First Fundamental Theorem of Asset Pricing

Theorem (Theorem 2)

A finite single-period financial market is **arbitrage-free (NA)** *if and only if there exists a probability measure P^* on $(\Omega, 2^\Omega)$ such that $P^*(\omega_k) > 0$ for every $\omega_k \in \Omega$, and:*

$$\mathbb{E}^* \left[\frac{\boldsymbol{S}_1}{1+r} \right] = \boldsymbol{S}_0$$

i.e., the discounted stock price process $\{\boldsymbol{S}_t(1+r)^{-t}\}_{t=0,1}$ is a vector-valued martingale under P^*.

Geometric Setup for the Proof ($n = 2$)

Let $\boldsymbol{S}_t = (S_{t,1}, S_{t,2})$ be two risky assets. Define the convex hull of discounted payoffs:

$$D := \text{conv} \left\{ \frac{\boldsymbol{S}_1(\omega_1)}{1+r}, \ldots, \frac{\boldsymbol{S}_1(\omega_N)}{1+r} \right\}$$

Two mutually exclusive possibilities exist: **(I)** $\boldsymbol{S}_0 \in D_0$ or **(II)** $\boldsymbol{S}_0 \notin D_0$.

Proof of Theorem 2: Steps 1 and 2

Step 1: $\exists$ EMM $\iff$ $S_0 \in D_0$

Directly from the definition of the relative interior D_0, setting $\alpha_k = p_k^* > 0$ gives:

$$S_0 = \sum_{k=1}^{N} p_k^* \frac{S_1(\omega_k)}{1+r} = \mathbb{E}^* \left[\frac{S_1}{1+r} \right]$$

Step 2: $\exists$ EMM $\implies$ NA

Prove by contradiction. Suppose an EMM exists but there is an arbitrage opportunity (Δ, b) such that $V_0 = \Delta \cdot S_0 + b = 0 \implies b = -\Delta \cdot S_0$. The terminal value satisfies:

$$V_1(\omega) = \Delta \cdot S_1(\omega) + b(1+r) = (1+r)\Delta \cdot \left(\frac{S_1(\omega)}{1+r} - S_0 \right) \geq 0 \ (\forall \omega), \text{ and } > 0 \text{ for atleat one } \omega$$

Taking expectations under the EMM P^*:
$\mathbb{E}^* \left[\frac{V_1}{1+r} \right] = \mathbb{E}^* \left[\Delta \cdot \frac{S_1}{1+r} - \Delta \cdot S_0 \right] = \Delta \cdot \mathbb{E}^* \left[\frac{S_1}{1+r} \right] - \Delta \cdot S_0 = 0$ However, since $V_1(\omega) \geq 0, \ \forall \omega$ and $V_1(\omega_k) > 0$ for at least one state with $p_k^* > 0$, $\mathbb{E}^* \left[V_1 (1+r)^{-1} \right] > 0$, yielding a contradiction.

Proof of Theorem 2: Step 3 (NA $\implies$ $\boldsymbol{S}_0 \in D_0$)

Proof by Contradiction.

Suppose NA holds but condition **(II)** is true, meaning $\boldsymbol{S}_0 \notin D_0$.

By the **Separation Theorem**, there exists a straight line/hyperplane L separating $\boldsymbol{S}_0$ from D_0. Denoting the normal vector by $\boldsymbol{\Delta} := (\Delta_1, \Delta_2)$ pointing into D_0, we have:

$$\boldsymbol{\Delta} \cdot (\boldsymbol{x} - \boldsymbol{S}_0) > 0, \quad \forall \boldsymbol{x} \in D_0$$

Since the vertices $\frac{\boldsymbol{S}_1(\omega)}{1+r}$ lie in the closure of D_0, this implies:

$$\boldsymbol{\Delta} \cdot \left(\frac{\boldsymbol{S}_1(\omega)}{1+r} - \boldsymbol{S}_0 \right) \begin{cases} \geq 0 & \forall \omega \in \Omega \\ > 0 & \text{for at least one } \omega \end{cases}$$

This constructs an explicit **arbitrage opportunity**, contradicting our initial assumption of NA. Thus, NA $\implies$ $\boldsymbol{S}_0 \in D_0$ $\implies$ $\exists$ EMM. □

The Arbitrage Pricing Theorem

Theorem (Theorem 3)

In a single-period arbitrage-free market, the discounted value process $\{V_t(1+r)^{-t}\}_{t=0,1}$ *of any trading strategy* $(\boldsymbol{\Delta}, b)$ *is a martingale under any EMM* P^**:*

$$\mathbb{E}^*\left[\frac{V_1}{1+r}\right] = V_0$$

Valuation of Attainable Claims

A contingent claim X is **attainable** (replicable) if there exists a self-financing strategy $(\boldsymbol{\Delta}, b)$ such that $V_1(\omega) = X(\omega)$ for all $\omega \in \Omega$. Such a strategy is called a **hedging strategy**.

Its unique arbitrage-free price at $t = 0$ is given by:

$$X^* := V_0 = \mathbb{E}^*\left[\frac{X}{1+r}\right]$$

Crucially, the right-hand side is invariant to the specific choice of EMM P^* if multiple exist.

Market Completeness and EMM Uniqueness

Definition (Complete Market)

A financial market is **complete** if every contingent claim X is attainable (can be perfectly replicated by a trading portfolio).

Theorem (Theorem 4: The Completeness Theorem)

An arbitrage-free market is complete if and only if there exists a **unique** *Equivalent Martingale Measure (EMM) P^*.*

Observation 1: The Linear System for EMMs

Finding an EMM requires solving a system of equations for $p_1^*, \ldots, p_N^*$:

$$\begin{aligned} p_1^* S_1^1(\omega_1) + \cdots + p_N^* S_1^1(\omega_N) = (1+r)S_0^1, \cdots, p_1^* S_1^n(\omega_1) + \cdots + p_N^* S_1^n(\omega_N) &= (1+r)S_0^n \\ p_1^* + \cdots + p_N^* = 1 & \end{aligned}$$

For a unique solution to exist, the market must have $n+1 = N$ assets and a non-singular system matrix A ($\det A \neq 0$).

Proof Sketch of the Completeness Theorem

Observation 2: Replicating an Arbitrary Claim X

A claim $X(\omega)$ is replicated if we can find a portfolio $(\boldsymbol{\Delta}, b)$ solving the N linear equations:

$$\Delta_1 S_1^1(\omega_k) + \cdots + \Delta_n S_1^n(\omega_k) + b(1+r) = X(\omega_k), \quad \forall k = 1, \ldots, N$$

The system matrix for replication is A^T.

Proof Sketch ($\Leftarrow$ and $\Rightarrow$).

($\Leftarrow$) If the EMM is unique, then $n+1 = N$ and the system matrix A is non-singular ($\det A \neq 0$). Consequently, A^T is also non-singular, meaning a unique replicating strategy $(\boldsymbol{\Delta}, b)$ exists for **any** arbitrary claim X. Thus, the market is complete.

($\Rightarrow$) Assume the market is complete. Suppose there exist two distinct EMMs P^* and $\tilde{P}^*$. By Theorem 3, for any claim X, $\mathbb{E}^*[\frac{X}{1+r}] = \tilde{\mathbb{E}}^*[\frac{X}{1+r}]$. By choosing indicator claims $X_i(\omega_k) = \delta_{ik}$, we get:

$$p_i^* = \sum_{k=1}^{N} p_k^* \delta_{ik} = \mathbb{E}^*[X_i] = \tilde{\mathbb{E}}^*[X_i] = \tilde{p}_i^* \implies P^* = \tilde{P}^* \quad \text{(Uniqueness)}$$

Multiperiod Binomial market:

One risky asset with price S_t

$$S_{t-1} \begin{cases} \nearrow S_t = u S_{t-1} \\ \searrow S_t = d S_{t-1} \end{cases} \qquad B_t = (1+r)^t$$

trading strategy (Δ_t, b_t) , $t = 1, 2, \ldots, T$

Δ_t: shares held during $(t-1, t]$

b_t: bond held during $(t-1, t]$

with a property $\Delta_t = \Delta_t(S_0, \ldots, S_{t-1})$

$b_t = b_t(S_0, \ldots, S_{t-1})$, $t = 1, 2, \ldots, T$

trading strategy $\{(\Delta_t, b_t)\}$ is called self-financing if

$$\Delta_t S_t + b_t (1+r)^t = \Delta_{t+1} S_t + b_{t+1} (1+r)^t \longrightarrow (1)$$

$t = 1, 2, \ldots, T-1$

$t = 0$, $V_0 = \Delta_1 S_0 + b_1$

$t = 1$, $V_1 = \Delta_1 S_1 + b_1 (1+r) \overset{\text{self financing}}{=} \Delta_2 S_1 + b_2 (1+r)$

resupply (Δ_1, b_1) to become (Δ_2, b_2)

To find the fair price of a given claim $X = g(S_T)$ we want to exploit the same replication idea:

Construct a self-financing portfolio $\{(\Delta_t, b_t)\}$ with value process $V_t = \Delta_t S_t + b_t (1+r)^t$, $t = 1, 2, \ldots, T$ and $V_0 = \Delta_1 S_0 + b_1$ in agreement with (1)

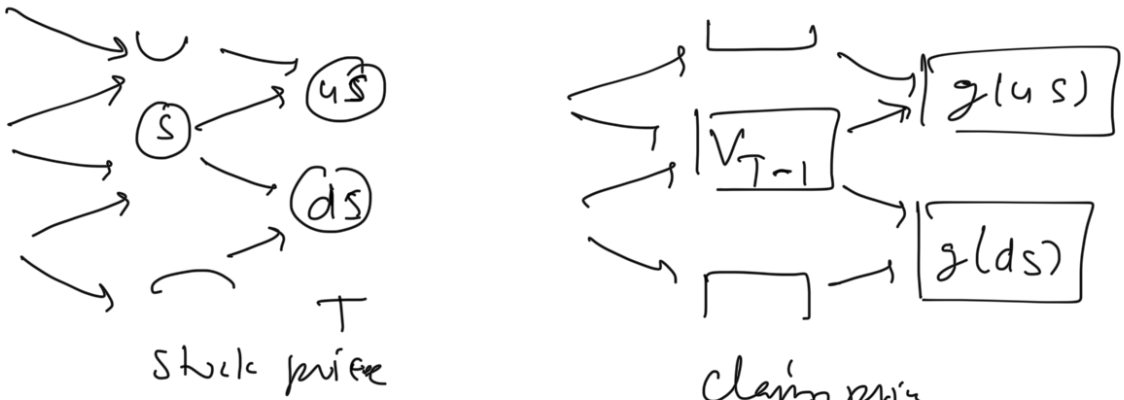
whose time $t=1$ value would exactly replicate the claim, i.e.

$$V_T(\omega) \equiv \Delta_T(\omega) S_t(\omega) + b_T(\omega)(1+r)^T$$
$$= X(\omega), \quad \omega \in \Omega$$

$$d < 1+r < u \quad , \quad X = g(S_T)$$

Suppose $t = T-1$, $S_{T-1} = s$

$t = T$ stock price us or ds



T^{th} time period

$$\Delta_T(s) = \frac{g(us) - g(ds)}{us - ds} = \frac{V_T[us] - V_T[ds]}{us - ds}$$

$$b_T(s) = \frac{u g(ds) - d g(us)}{(1+r)^T (u-d)} = \frac{u V_T[ds] - d V_T[us]}{(1+r)^T (u-d)}$$

$t = T-1$

$$V_{T-1}[s] = \frac{1}{1+r}\left[p^* g(us) + (1-p^*) g(ds)\right]$$
$$= E^*\left(\frac{g(S_T)}{1+r} \,\middle|\, S_{T-1} = s\right),$$

where $p^* = \frac{1+r-d}{u-d} \in (0,1)$

We move one step back and repeat the Ex.

to find $\Delta_{T-1}[s]$, $b_{T-1}[s]$ and $V_{T-2}[s]$

$t = T-2$, stock price is s

$$\Delta_{T-1}[s] = \frac{V_{T-1}[us] - V_{T-1}[ds]}{us - ds}$$

$$b_{T-1}[s] = \frac{u V_{T-1}[ds] - d V_{T-1}[us]}{(1+r)^{T-1}(u-d)}$$

and $V_{T-2}[s] = E^*\left[\frac{V_{T-1}[S_{T-1}]}{1+r} \mid S_{T-2} = s\right]$

$$= \frac{1}{1+r}\left[p^* V_{T-1}[us] + (1-p^*) V_{T-1}[ds]\right]$$

$$= \frac{1}{(1+r)^2} \sum_{j=0}^{2} \binom{2}{j} (p^*)^j (1-p^*)^{2-j} g(u^j d^{2-j} s)$$

$$\vdots$$

General formula for $t = 1, 2, \ldots T$

$$\Delta_t[s] = \frac{V_t[us] - V_t[ds]}{us - ds}, \quad b_t[s] = \frac{u V_t[ds] - d V_t[us]}{(1+r)^t (u-d)}$$

and

$$V_t[s] = \frac{1}{(1+r)^{T-t}} \sum_{j=0}^{T-t} \binom{T-t}{j} (p^*)^j (1-p^*)^{T-t-j} g(u^j d^{T-t-j} s)$$

$$S_t = E^*\left(\frac{S_{t+1}}{1+r} \mid S_t = s_t\right)$$

$$= E^*\left(\frac{S_{t+1}}{1+r} \mid S_0 = s_0, \dots, S_t = s_t\right)$$

Under p^*, the discounted stock price process

$S_t^* = S_t (1+r)^{-t}$ is MG

$E(S_{t+1}^* \mid S_0^*, \dots, S_t^*) = S_t^*, \quad t = 0, 1, \dots, T-1$

Like in single-period case $\underset{\sim}{p}^*$ is the only prob. measure on the sample space

$\Omega = \{\omega = (v_1, \dots, v_T) : v_T = u \text{ or } d, \ t = 1, \dots, T\}$

under which S_t^* is MG

$t = 0$ general formula for claim value

$$V_0[S_0] = \frac{1}{(1+r)^T} \sum_{j=0}^{T} \binom{T}{j} (p^*)^j (1-p^*)^{T-j} g(u^j d^{T-j} S_0)$$

$$= E^*\left(\frac{g(S_0 e^{Z_T})}{(1+r)^T}\right)$$

$Z_n := Y_1 + \dots + Y_n, \quad n \geq 0$

is a random walk with IID jumps

$$Y_k = \begin{cases} \ln u & \text{wp } p^* \\ \ln d & \text{wp } 1-p^* \end{cases}$$

$S_n = S_0 e^{Z_n}$

$\frac{S_n}{S_0} = e^{Z_n}$

$V_0[S_0] = \frac{1}{(1+r)^n}$

$$\gamma_n\left(\frac{\bar{x}_n}{s_0}\right) = z_n = \sum_{i=1} y_i$$

Example : Pricing European call in Binomial setting

$t=0$, price of European call is

$$C = \frac{1}{(1+r)^T} \sum_{j=0}^{T} \binom{T}{j} (p^*)^j (1-p^*)^{T-j} (u^j d^{T-j} S_0 - k)^+$$

$$= \frac{1}{(1+r)^T} \sum_{j=m}^{T} \binom{T}{j} (p^*)^j (1-p^*)^{T-j} (u^j d^{T-j} S_0 - k)$$

where

$$m := \left\lfloor \frac{\ln k - \ln S_0 - T \ln d}{\ln u - \ln d} \right\rfloor + 1$$

$$u^j d^{T-j} S_0 > k$$
$$\ln S_0 + j \ln u + (T-j) \ln d > \ln k$$
$$j(\ln u - \ln d) > \ln k - \ln S_0 - T \ln d$$

$$C = S_0 \sum_{j=m}^{T} \binom{T}{j} \underbrace{\left(\frac{u p^*}{1+r}\right)^j}_{=\hat{p}} \underbrace{\left(\frac{d p^*}{1+r}\right)^{T-j}}_{\to 1-\hat{p}} - \frac{k}{(1+r)^T} \sum_{j=m}^{T} \binom{T}{j} (p^*)^j (1-p^*)^{T-j}$$

$$= S_0 P(\hat{U} \geq m) - \frac{k}{(1+r)^T} P(U^* \geq m) \quad \text{✡}$$

$\hat{U} \sim B(n, \hat{p})$, $U^* \sim B(n, p^*)$

, $n = 0, 1, \dots, T$

—x—

$[0, T]$ split into larger time period of length $\delta_n := \frac{T}{n}$

$n \to \infty$

$t = 0, \delta_n, 2\delta_n, \dots, n\delta_n$

Binomial model choose $u = u_n$, $d = d_n$, $r = r_n$ as $n \to \infty$

fixed maturity time $T > 0$

$[(j-1)\delta_n, j\delta_n]$, $j = 1, \dots, n$

$$S_{(j-1)\delta_n} \begin{cases} \nearrow S_{j\delta_n} = u_n S_{(j-1)\delta_n} \\ \searrow S_{j\delta_n} = d_n S_{(j-1)\delta_n} \end{cases} \quad , j = 1, \to n$$

$$u_n = e^{\sigma\sqrt{\delta_n}} \simeq 1 + \sigma\sqrt{\delta_n}$$

$$d_n = e^{-\sigma\sqrt{\delta_n}} \simeq 1 - \sigma\sqrt{\delta_n}$$

$$r_n = e^{r\delta_n} - 1 \simeq r\delta_n$$

$$S_n = S_0 \, e^{\sum_{i=1}^{n} \gamma_i} = S_0 \, e^{Z_n}$$

$Z_n = \sum_{i=1}^{n} \gamma_i$ is a random walk with i.i.d. jumps

$$\gamma_{n,k} = \begin{cases} \ln u_n = \sigma\sqrt{\frac{T}{n}} & \text{w.p. } p_n^* \\ \ln d_n = -\sigma\sqrt{\frac{T}{n}} & \text{w.p. } 1 - p_n^* \end{cases}$$

with $p^* = \dfrac{1 + r_n - d_n}{u_n - d_n} = \dfrac{e^{r\delta_n} - e^{-\sigma\sqrt{\delta_n}}}{e^{\sigma\sqrt{\delta_n}} - e^{-\sigma\sqrt{\delta_n}}}$

$n \to \infty$, time $t = 0$, price of European call is

$$C = S_0 \, P(U^* \geq m) - \frac{k}{(1+r)^T} P(U^* \geq m)$$

where $U^* \sim$

$$= S_0 \left(1 - \Phi\left(\frac{m - n\hat{p}_n}{\sqrt{n\hat{p}_n(1-\hat{p}_n)}}\right)\right) - \frac{k}{(1+r)^T}\left(1 - \Phi\left(\frac{m - np_n^*}{\sqrt{np_n^*(1-p_n^*)}}\right)\right)$$

$$= S_0 \, \Phi\left(\frac{n\hat{p}_n - m}{\sqrt{n\hat{p}_n(1-\hat{p}_n)}}\right) - \frac{k}{(1+r)^T} \, \Phi\left(\frac{np_n^* - m}{\sqrt{np_n^*(1-p_n^*)}}\right)$$

$$p_n^* = \frac{1 + r_n - d_n}{u_n - d_n} = \frac{e^{r s_n} - e^{-\sigma\sqrt{s_n}}}{e^{\sigma\sqrt{s_n}} - e^{-\sigma\sqrt{s_n}}}$$

$$= \frac{1 + r\frac{T}{n} - 1 + \sigma\sqrt{\frac{T}{n}}}{1 + \sigma\sqrt{\frac{T}{n}} - 1 + \sigma\sqrt{\frac{T}{n}}} = \frac{r\sqrt{\frac{T}{n}} + \sigma}{2\sigma}$$

$$= \frac{1}{2} + \frac{r}{\sigma}\sqrt{\frac{T}{n}}$$

$$\hat{p}_n = \frac{u_n p_n^*}{1 + r_n}$$

$$C = S_0 \, \Phi(h) - k e^{-rT} \, \Phi(h - \sigma\sqrt{T}),$$

$$\text{where } h := \frac{\ln(S_0/k) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

which gives us Black-Scholes formula giving the price of European call with maturity T and strike price k under the assumption of cont.-time financial market with the

stock modelled by geometric BM

$S_t = S_0 e^{\mu t + \sigma W_t}$, bond price $B_t = e^{rt}$,

$t \in [0, T]$

Multiperiod Binomial Market and the Black-Scholes Formula

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Multiperiod Binomial Market Setup

Consider a market with a risky asset S_t and a risk-free asset B_t over time periods $t = 1, 2, \ldots, T$:

- **Risky Asset (S_t):**

$$S_t = \begin{cases} uS_{t-1} & \text{with probability } p^* \\ dS_{t-1} & \text{with probability } 1 - p^* \end{cases}$$

- **Risk-Free Asset (B_t):**

$$B_t = (1 + r)^t$$

Trading Strategy: (Δ_t, b_t) held during $(t-1, t]$, where:

- $\Delta_t = \Delta_t(S_0, \ldots, S_{t-1})$ is the number of stock shares.
- $b_t = b_t(S_0, \ldots, S_{t-1})$ is the amount in the risk-free bond.

Self-Financing Trading Strategies

A trading strategy $\{(\Delta_t, b_t)\}$ is called **self-financing** if the portfolio value process satisfies:

$$\Delta_t S_t + b_t(1+r)^t = \Delta_{t+1} S_t + b_{t+1}(1+r)^t \quad \text{for } t = 1, 2, \ldots, T-1$$

- **At** $t = 0$**:** $V_0 = \Delta_1 S_0 + b_1$
- **At** $t = 1$**:** $V_1 = \Delta_1 S_1 + b_1(1+r) \overset{\text{self-fi}}{=} \Delta_2 S_1 + b_2(1+r)$

To find the **fair price** of a contingent claim $X = g(S_T)$, we construct a self-financing portfolio that replicates the claim exactly at maturity:

$$\text{Find } \{(\Delta_t, b_t)\} \quad \text{such that } V_T(\omega) = X(\omega) \quad \forall \omega \in \Omega$$

Replication Strategy via Backward Induction

Assuming $d < 1 + r < u$, consider the final period from $t = T - 1$ to $t = T$ given $S_{T-1} = s$:

$$\Delta_T(s) = \frac{g(us) - g(ds)}{us - ds} = \frac{V_T[us] - V_T[ds]}{us - ds}$$

$$b_T(s) = \frac{ug(ds) - dg(us)}{(1+r)^T(u-d)} = \frac{uV_T[ds] - dV_T[us]}{(1+r)^T(u-d)}$$

At time $t = T - 1$, the portfolio value is given by the risk-neutral expectation:

$$V_{T-1}[s] = \frac{1}{1+r}\left[p^* g(us) + (1-p^*)g(ds)\right] = E^*\left(\frac{g(S_T)}{1+r}\,\middle|\, S_{T-1} = s\right)$$

Where the **risk-neutral probability** is $p^* = \frac{1+r-d}{u-d} \in (0, 1)$.

General Formula and Martingale Property

Moving backward step-by-step for $t = 1, 2, \ldots, T$, we get the general replicating parameters:

$$\Delta_t[s] = \frac{V_t[us] - V_t[ds]}{us - ds}, \quad b_t[s] = \frac{uV_t[ds] - dV_t[us]}{(1+r)^t(u-d)}$$

The general value process formula at any time t is:

$$V_t[s] = \frac{1}{(1+r)^{T-t}} \sum_{j=0}^{T-t} \binom{T-t}{j} (p^*)^j (1-p^*)^{T-t-j} g(u^j d^{T-t-j} s)$$

Under the risk-neutral measure p^*, the discounted stock price process $S_t^* = S_t(1+r)^{-t}$ is a **Martingale (MG)**:

$$E^*(S_{t+1}^* \mid S_0^*, \ldots, S_t^*) = S_t^*$$

Pricing a European Call Option

For a European Call option with strike price K, the claim is $g(S_T) = (S_T - K)^+$. At time $t = 0$, the fair price C is:

$$C = \frac{1}{(1+r)^T} \sum_{j=0}^{T} \binom{T}{j} (p^*)^j (1-p^*)^{T-j} (u^j d^{T-j} S_0 - K)^+$$

Let m be the minimum number of upward moves needed for the option to finish in-the-money ($S_T > K$):

$$m = \left\lfloor \frac{\ln K - \ln S_0 - T \ln d}{\ln u - \ln d} \right\rfloor + 1$$

This simplifies to two distinct binomial probabilities:

$$C = S_0 P(\hat{U} \geq m) - \frac{K}{(1+r)^T} P(U^* \geq m)$$

Where $U^* \sim \text{Bin}(T, p^*)$ and $\hat{U} \sim \text{Bin}(T, \hat{p})$ with $\hat{p} = \frac{up^*}{1+r}$.

Taking the Continuous Limit

Divide the time horizon $[0, T]$ into n layers of length $\delta_n = \frac{T}{n}$. As $n \to \infty$:

$$u_n = e^{\sigma\sqrt{\delta_n}} \simeq 1 + \sigma\sqrt{\delta_n}$$
$$d_n = e^{-\sigma\sqrt{\delta_n}} \simeq 1 - \sigma\sqrt{\delta_n}$$
$$r_n = e^{r\delta_n} - 1 \simeq r\delta_n$$

The stock price becomes a random walk with independent, identically distributed jumps:

$$S_n = S_0 e^{Z_n} = S_0 e^{\sum_{i=1}^{n} Y_i}$$

The risk-neutral probability limits to:

$$p_n^* = \frac{e^{r\delta_n} - e^{-\sigma\sqrt{\delta_n}}}{e^{\sigma\sqrt{\delta_n}} - e^{-\sigma\sqrt{\delta_n}}} \to \frac{1}{2} + \frac{r}{2\sigma}\sqrt{\delta_n}$$

The Black-Scholes Formula

By applying the Central Limit Theorem to the binomial probabilities as $n \to \infty$, the discrete distribution converges to a standard normal distribution $\Phi(\cdot)$:

$$C = S_0 \Phi(h) - K e^{-rT} \Phi(h - \sigma\sqrt{T})$$

Where:

$$h := \frac{\ln(S_0/K) + \left(r + \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}}$$

This matches the continuous-time framework modeled by geometric Brownian motion:

$$S_t = S_0 e^{\mu t + \sigma W_t}, \quad B_t = e^{rt} \quad \text{for } t \in [0, T]$$

Martingale and Claim pricing:

price process $(S_t := S_t^1, \dots, S_t^n)_{t \geq 0}$ is a S.P. on $(\Omega, \mathcal{F}, P)$ — n diff. (risky) assets

bond / bank account

$$B_t = \begin{cases} (1+r)^t & \text{in discrete time } t = 0, 1, 2, \dots \\ e^{rt} & \text{" cont time } t \geq 0 \end{cases}, \quad r \geq 0$$

European contingent claim $X \geq 0$ with maturity T

$g(S_t)$ with $t \leq T$

The "risk neutral" or "arbitrage free" price of claim is obtained as the value of the self financing portfolio that replicate the claim X at the maturity time T. Here are the three key facts of the no-arbitrage pricing theory

$\rightarrow$ $\exists$ an EMM P^* on $(\Omega, \mathcal{F})$

and SP $\underset{\sim}{S}_t^* = \begin{cases} (1+r)^{-t} \underset{\sim}{S}_t & \text{in discrete time} \\ e^{-rt} \underset{\sim}{S}_t & \text{" cont time} \end{cases}$ $t \geq 0$

is an MG on $(\Omega, \mathcal{F}, P^*)$, then the market is arbitrage free.

→ For an attainable claim X with maturity T at time t arbitrage-free price is given by

$$P_t(X) = E^*\left((1+r)^{t-T} X \mid \mathcal{F}_t\right) \quad \text{discrete}$$

$$E^*\left(e^{-r(T-t)} X \mid \mathcal{F}_t\right) \quad \text{cont}^t \; t \in [0,T]$$

$$P_0(X) = E^*\left((1+r)^{-T} X\right) \quad \text{or} \quad E^*\left(e^{-rT} X\right)$$

→ If the EMM P^* is unique, then market is complete, i.e., any claim is attainable and so can always be priced.

—X—

Black-Scholes Framework (cont time model)

bond $B_t = e^{rt}$, $t \in [0,T]$

$\left(dB_t = r B_t dt \;,\; B_0 = 1 \right)$

a risky asset (or stock) of which price S_t satisfying the Black-Scholes SDE

$$dS_t = \mu S_t dt + \sigma S_t dW_t \;, \quad t \in [0,T]$$

when $\mu \in \mathbb{R}$, $\sigma > 0$, $S_0 > 0$ const,

$\{W_t\}$ SBM/Wiener process

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t \quad \text{—— (1)}$$

Itô's formula

$$d(\ln S_t) = \frac{1}{S_t} dS_t - \frac{1}{2 S_t^2} (dS_t)^2$$

$$= \frac{dS_t}{S_t} - \frac{1}{2 S_t^2} \times \sigma^2 S_t^2 \, dt$$

$$= \mu \, dt + \sigma \, dW_t - \frac{\sigma^2}{2} dt$$

$$= \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma \, dW_t$$

$$\ln S_t - \ln S_0 = \left(\mu - \frac{\sigma^2}{2}\right) t + \sigma W_t$$

$$\Rightarrow S_t = S_0 \, e^{\left(\mu - \frac{\sigma^2}{2}\right) t + \sigma W_t}, \quad t \in [0, T]$$

—— ✡$_1$

Qs 1) Black-Scholes market arbitrage-free? If Yes, then one price attainable claims.

Q 2) Is the Black-Scholes market complete?

answers Q 1, 2 are given in terms of EMM

on $(\Omega, \mathcal{F})$ what probs measures are equivalent to

$\mathbb{P}_1$

Let $Z_t := S_0 \exp\left\{ \int_0^t a_s ds + \sigma W_t \right\}, \quad t \in [0, T]$

$\{a_t\}_{t \in (0,T)}$ an adapted process on $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$

Consider $\{e^{-rt} Z_t\}$ is an Itô process it will be MG

iff the coeff of dt-term in sto diff is zero.

Ex
Ito's process $X_t = X_0 + \int_0^t a_s ds + \int_0^t b_s dW_s$, $t \in [0,T]$ is an MG iff $a_t = 0$, $t \in [0,T]$

$$d(e^{-rt} Z_t) = d\Big(\underbrace{\exp\Big[\int_0^t a_s ds\Big]}_{=: X_t} \times \underbrace{\exp[-rt + \sigma W_t]}_{=: Y_t} \Big)$$

$$= Y_t\, dX_t + X_t\, dY_t + dX_t\, dY_t$$

$$= Y_t a_t X_t\, dt + X_t\Big(-r Y_t\, dt + \sigma Y_t\, dW_t + \frac{1}{2}\sigma^2 Y_t\, dt\Big) + 0$$

$$= X_t Y_t \Big(a_t - r + \frac{\sigma^2}{2}\Big) dt + \sigma X_t Y_t\, dW_t$$

As $X_t Y_t > 0$, the coeff of dt is zero iff $a_t = r - \frac{\sigma^2}{2}$, $t \in [0,T]$

∴ According to previous sections, the Black-Scholes market is arbitrage-free and complete and so one can price any contingent claim X in the market, using the valuation formula

$$P_t(X) = E^{\mathbb{Q}}\big(e^{-r(T-t)} X \mid \mathcal{F}_t\big)$$